

Caminho Mínimo (3)

Zenilton Patrocínio

Caminho Mínimo entre Todos os Vértices

O problema consiste em determinar os caminhos mínimos entre todos os pares de vértices do grafo.

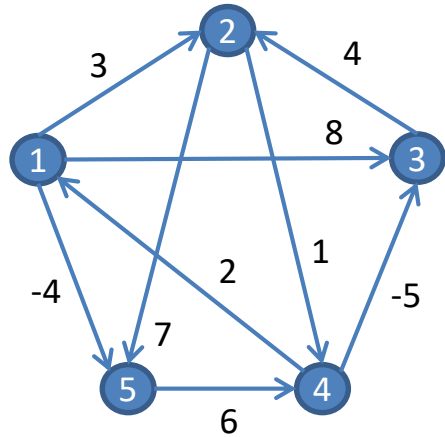
Se todos os custos forem positivos, para n vértices pode-se utilizar n repetições do método de Dijkstra → cada vez utiliza-se um vértice como raiz

- Custo → $n \times O(n^2) = O(n^3)$ para uma implementação trivial

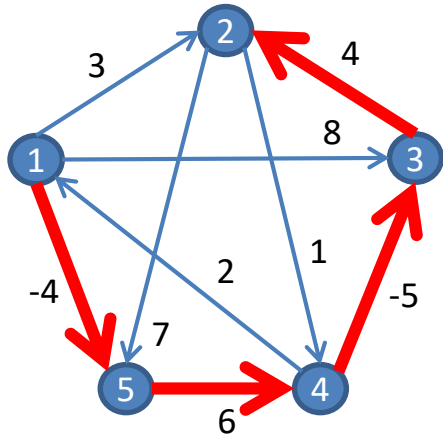
Se houver custos negativos (sem a presença de ciclos negativos), para n vértices pode-se utilizar n repetições do método de Bellman-Ford (com m arestas)

- Custo → $n \times O(nm) = O(n^2m) \rightarrow O(n^4) !!!$

Exemplo



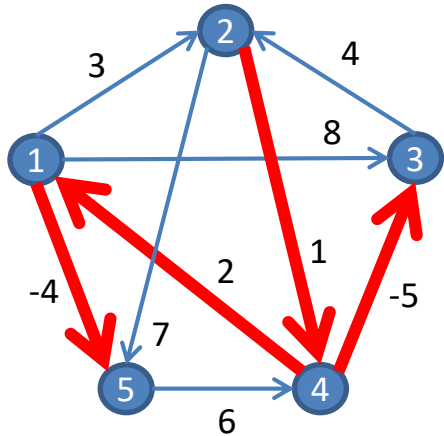
Exemplo



Distâncias Mínimas

	1	2	3	4	5
1	0	1	-3	2	-4

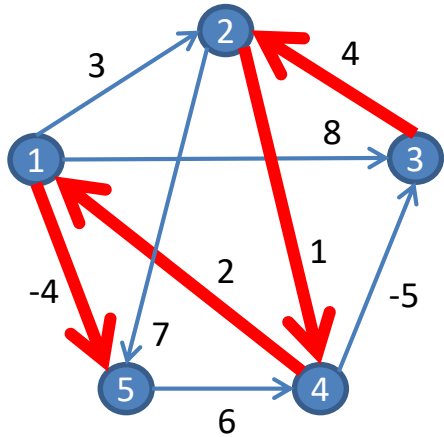
Exemplo



Distâncias Mínimas

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1

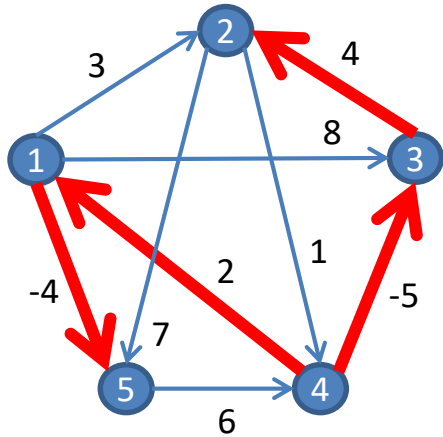
Exemplo



Distâncias Mínimas

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3

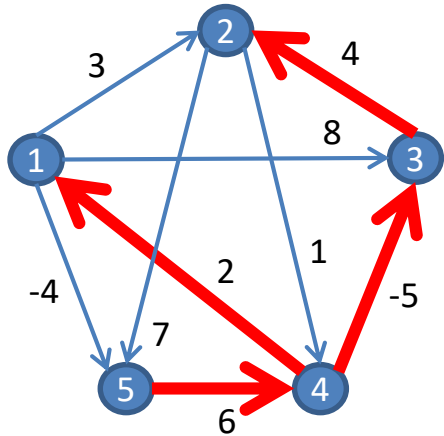
Exemplo



Distâncias Mínimas

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2

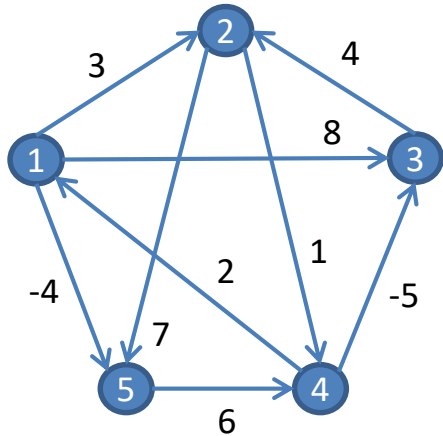
Exemplo



Distâncias Mínimas

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

Exemplo



Distâncias Mínimas

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
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4	2	-1	-5	0	-2
5	8	5	1	6	0

Método de Floyd-Warshall

Método de Floyd-Warshall

O algoritmo de Floyd-Warshall calcula os caminhos mais curtos entre todos os pares de vértices (consegue lidar arestas de peso negativo desde que não exista ciclo de custo negativo).

Trata-se novamente de um algoritmo de programação dinâmica.

Utiliza o conceito de **relaxação do comprimento dos caminhos** mais curtos de forma incremental aprimorando a estimativa de distância, até que o valor ótimo seja alcançado.

O algoritmo analisa os caminhos entre os vértices i e j passando por cada um dos k vértices intermediários possíveis, $k \in V(G)$.

Método de Floyd-Warshall

Considere que os vértices do grafo são numerados de 1 a n .

Seja $dist_{ij}^k$ a distância entre o par de vértices i e j usando como intermediários apenas os vértices do conjunto $\{1, 2, \dots, k\}$.

Em particular, $dist_{ij}^0 = d_{ij}$, se $\exists (i, j) \in E(G)$, $dist_{ij}^0 = \infty$, caso contrário. Além disso, $dist_{ii}^0 = 0$.

A relaxação do comprimento dos caminhos determina $dist_{ij}^k$ a partir das distâncias $dist_{**}^{k-1}$, isto é, utilizando um caminho de i para j que não passa por k ou o par de caminhos de i para k e de k para j (ambos não passando por k).

Método de Floyd-Warshall

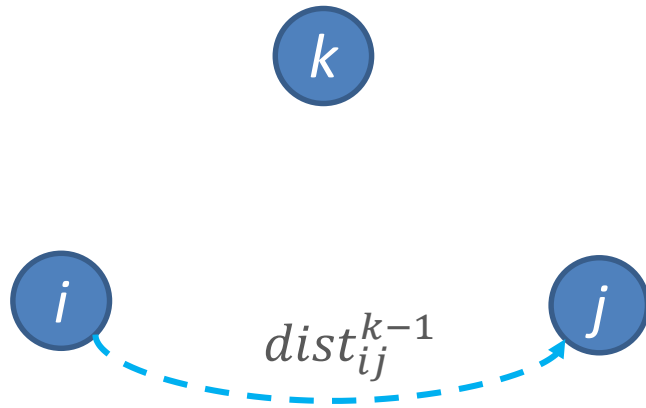
i

j

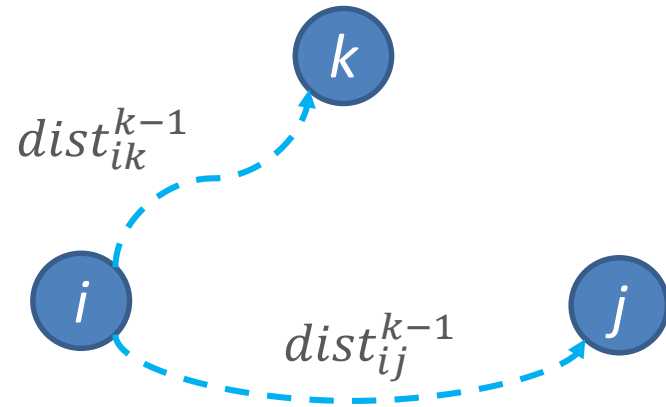
Método de Floyd-Warshall



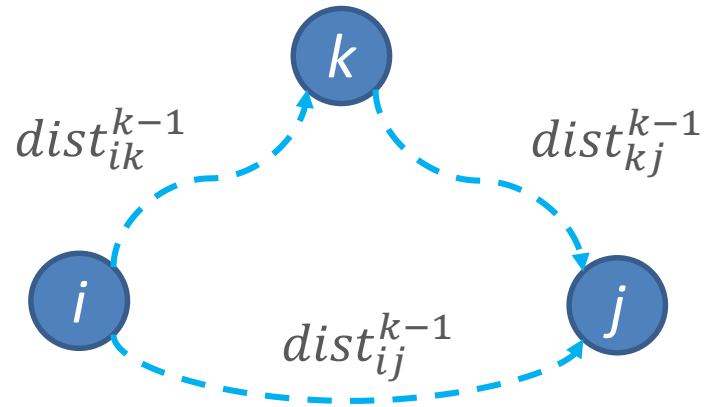
Método de Floyd-Warshall



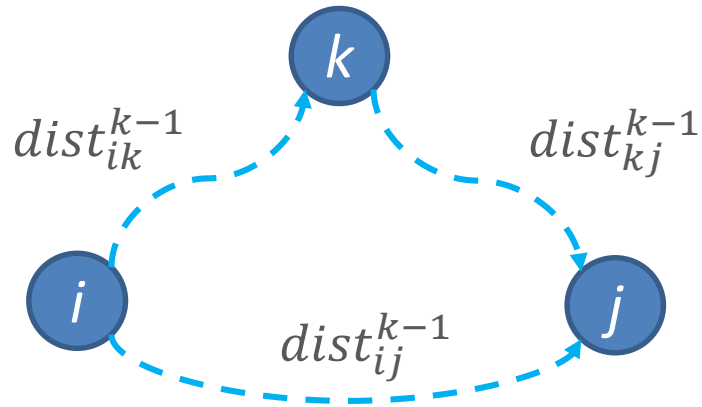
Método de Floyd-Warshall



Método de Floyd-Warshall

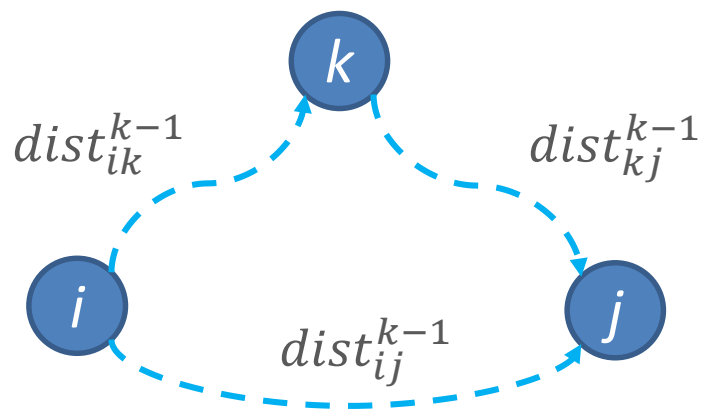


Método de Floyd-Warshall



$$dist_{ij}^k = \min\{dist_{ij}^{k-1}, dist_{ik}^{k-1} + dist_{kj}^{k-1}\}$$

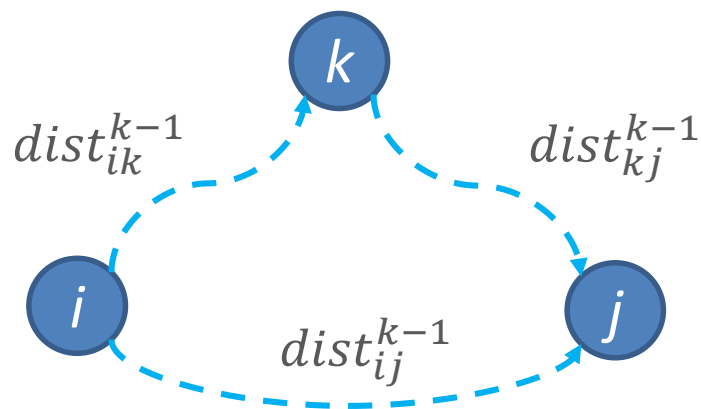
Método de Floyd-Warshall



se $dist[i,j] > dist[i,k] + dist[k,j]$ então
 $dist[i,j] \leftarrow dist[i,k] + dist[k,j]$

$$dist_{ij}^k = \min\{dist_{ij}^{k-1}, dist_{ik}^{k-1} + dist_{kj}^{k-1}\}$$

Método de Floyd-Warshall



$$dist_{ij}^k = \min\{dist_{ij}^{k-1}, dist_{ik}^{k-1} + dist_{kj}^{k-1}\}$$

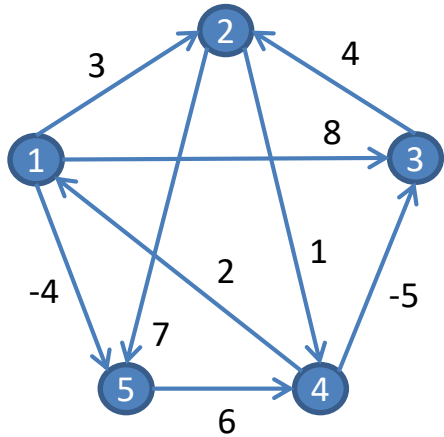
se $dist[i,j] > dist[i,k] + dist[k,j]$ então
 $dist[i,j] \leftarrow dist[i,k] + dist[k,j]$

Para $k = 1, 2, \dots, n$

Método de Floyd-Warshall – Algoritmo

1. para $i = 1, \dots, n$ faça // Inicializar distâncias
 para $j = 1, \dots, n \mid j \neq i$ faça $\text{dist}[i, j] \leftarrow \infty$;
 $\text{dist}[i, i] \leftarrow 0$;
2. para toda aresta $(i, j) \in E(G)$ faça $\text{dist}[i, j] \leftarrow d_{ij}$;
3. para $k = 1, \dots, n$ faça // Para cada possível intermediário
 para $i = 1, \dots, n$ faça
 para $j = 1, \dots, n$ faça
 se $\text{dist}[i, j] > \text{dist}[i, k] + \text{dist}[k, j]$ então // Testar caminho de i para j
 $\text{dist}[i, j] \leftarrow \text{dist}[i, k] + \text{dist}[k, j]$; // Atualizar a distância

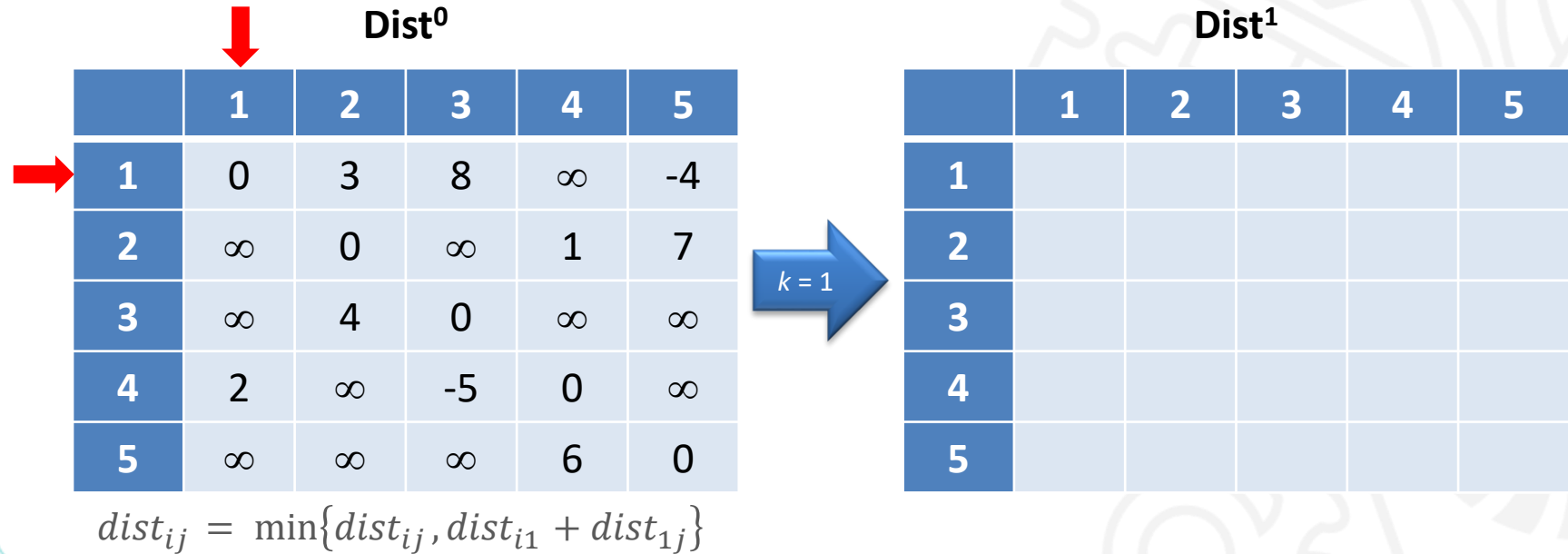
Método de Floyd-Warshall – Exemplo



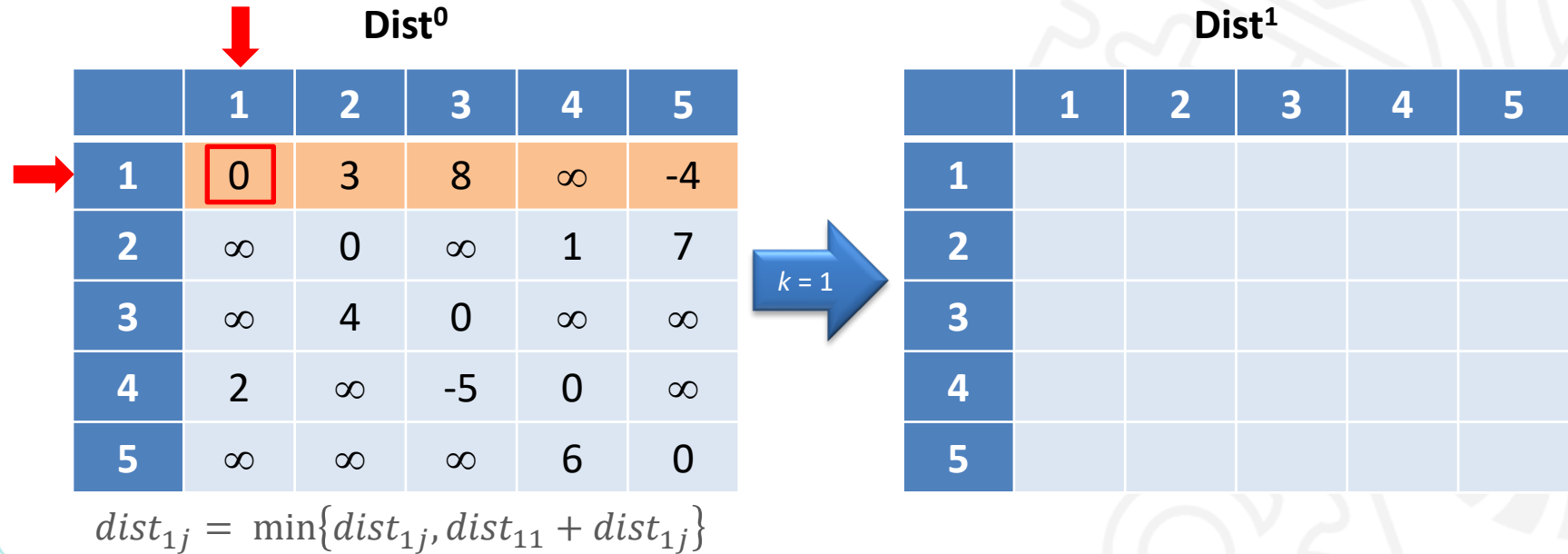
Dist⁰

	1	2	3	4	5
1	0	3	8	∞	-4
2	∞	0	∞	1	7
3	∞	4	0	∞	∞
4	2	∞	-5	0	∞
5	∞	∞	∞	6	0

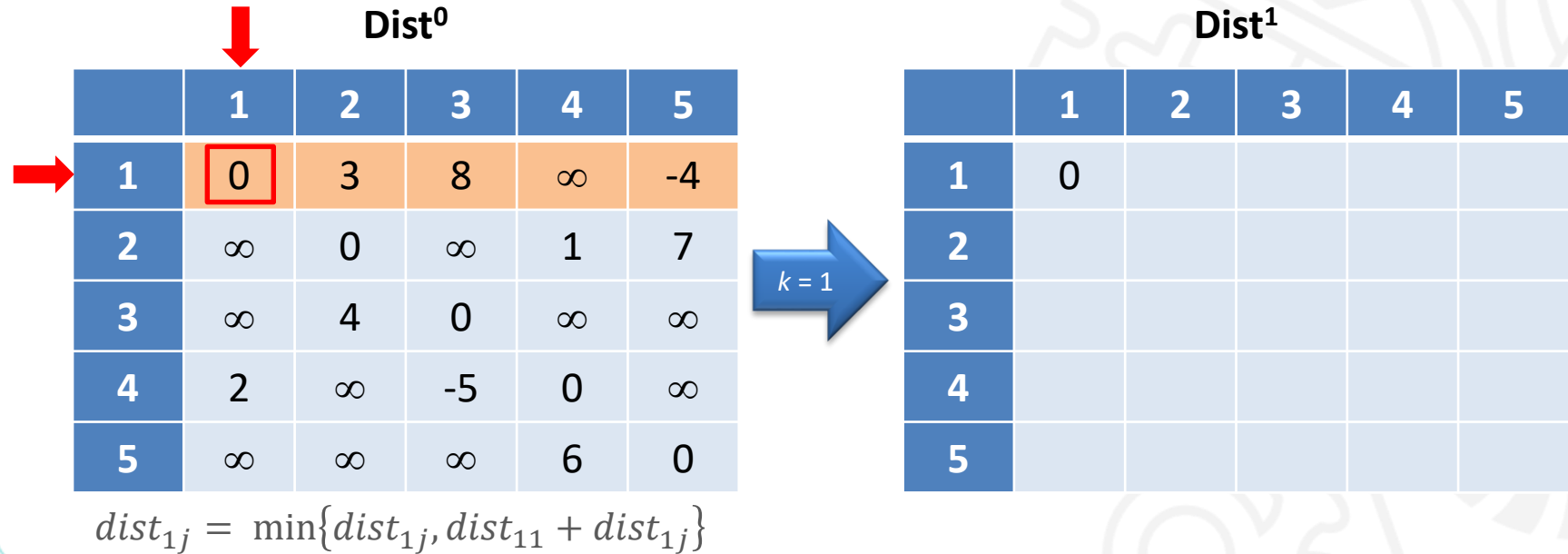
Método de Floyd-Warshall – Exemplo



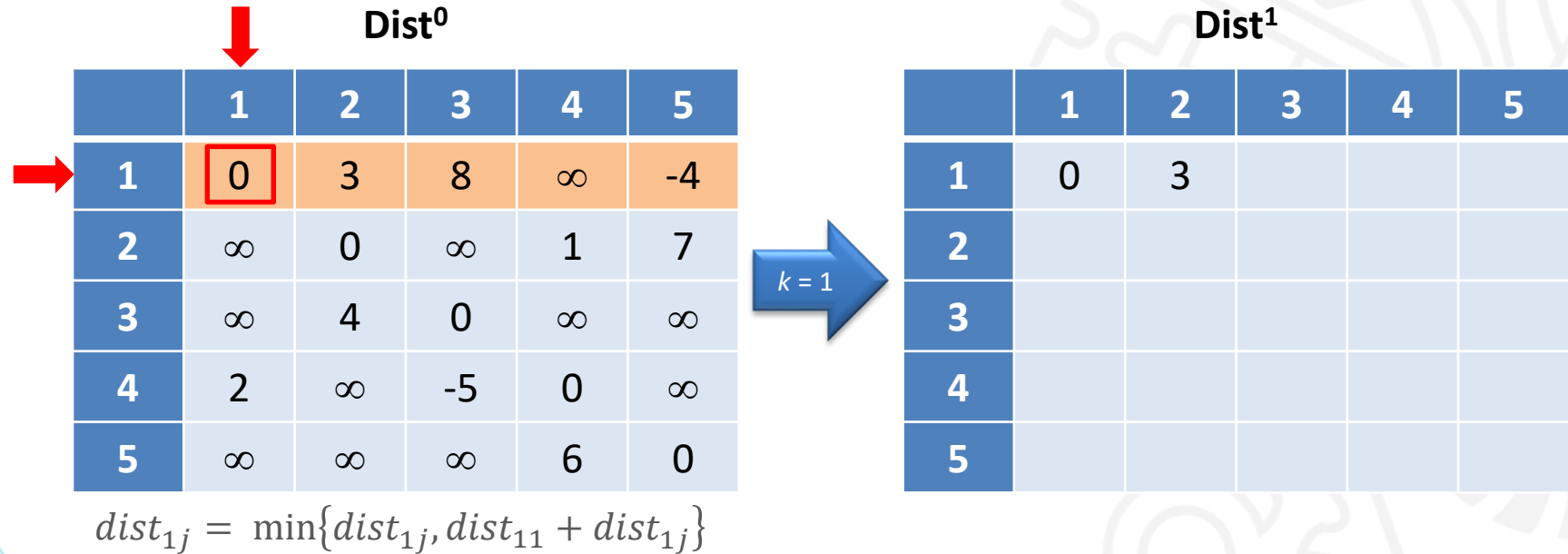
Método de Floyd-Warshall – Exemplo



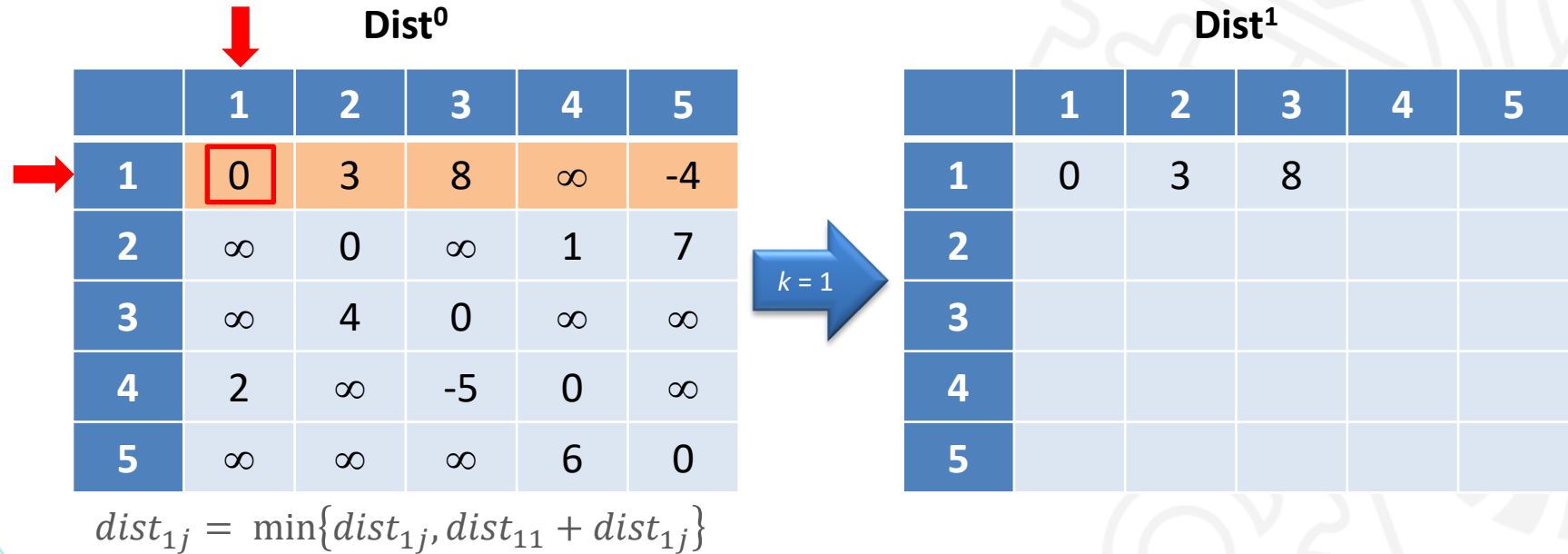
Método de Floyd-Warshall – Exemplo



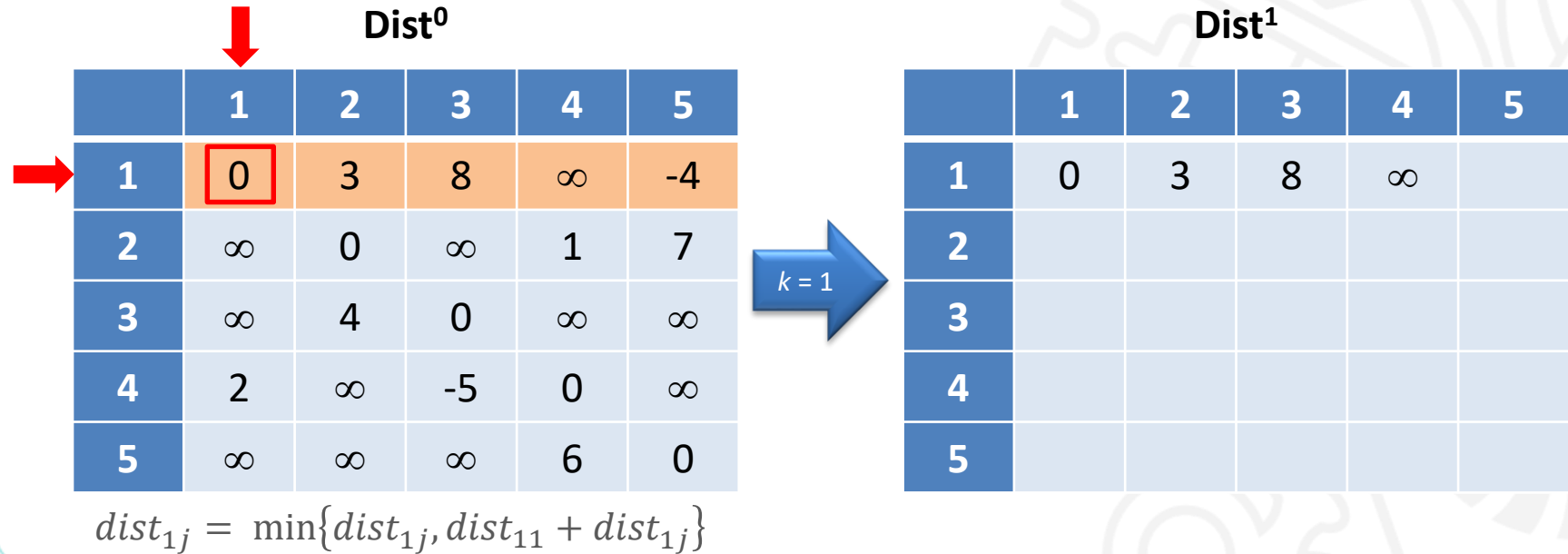
Método de Floyd-Warshall – Exemplo



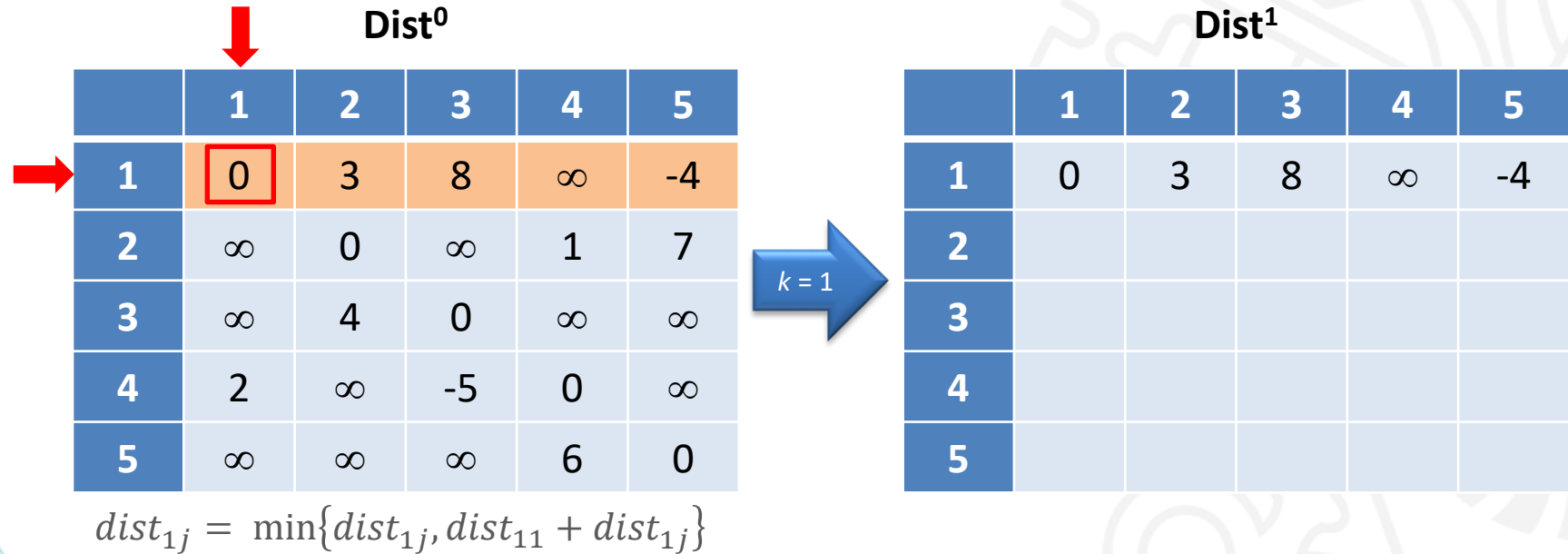
Método de Floyd-Warshall – Exemplo



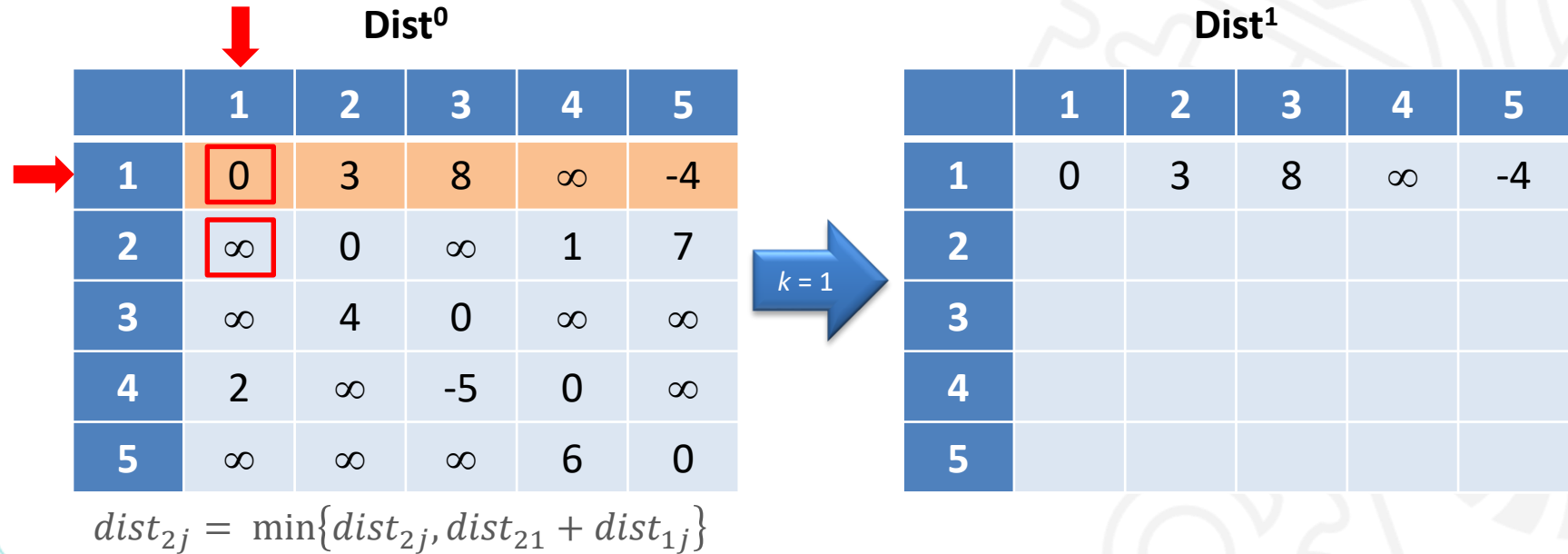
Método de Floyd-Warshall – Exemplo



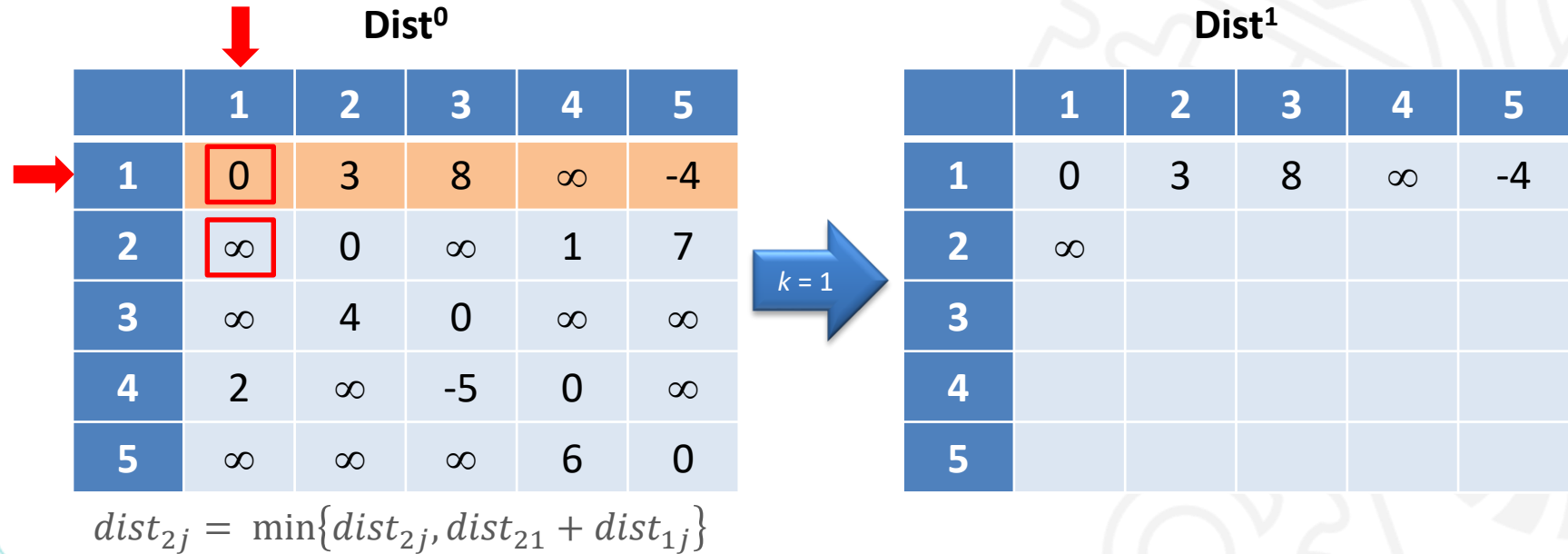
Método de Floyd-Warshall – Exemplo



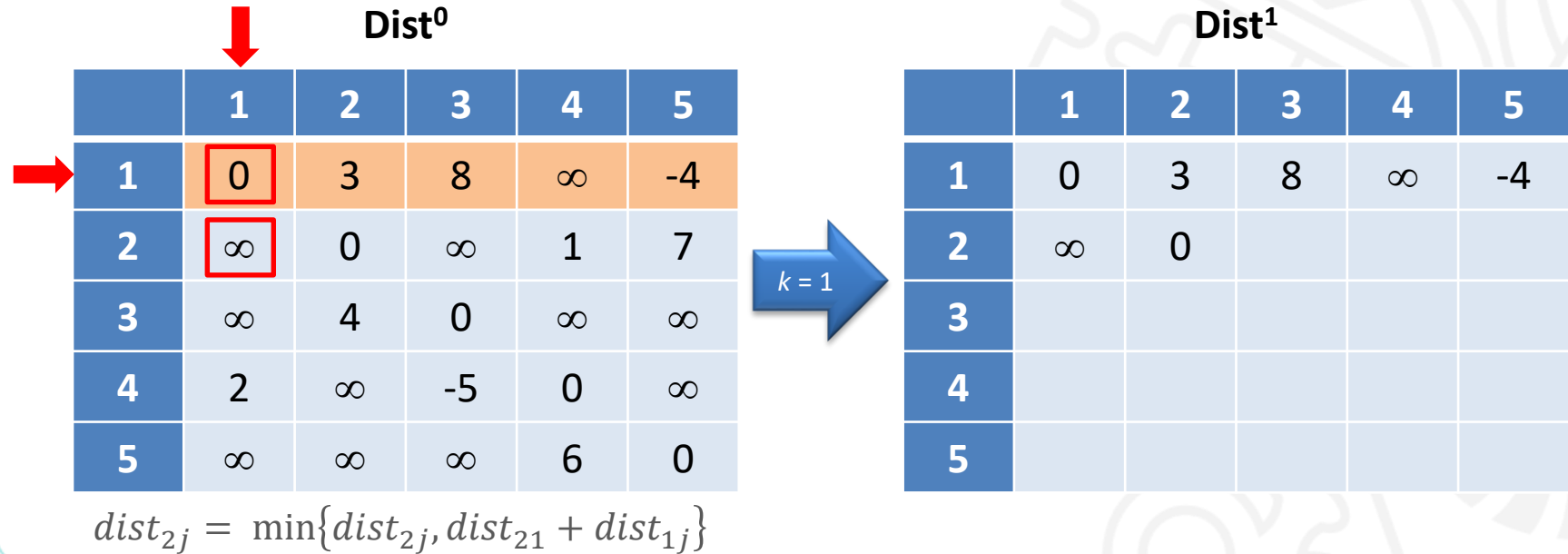
Método de Floyd-Warshall – Exemplo



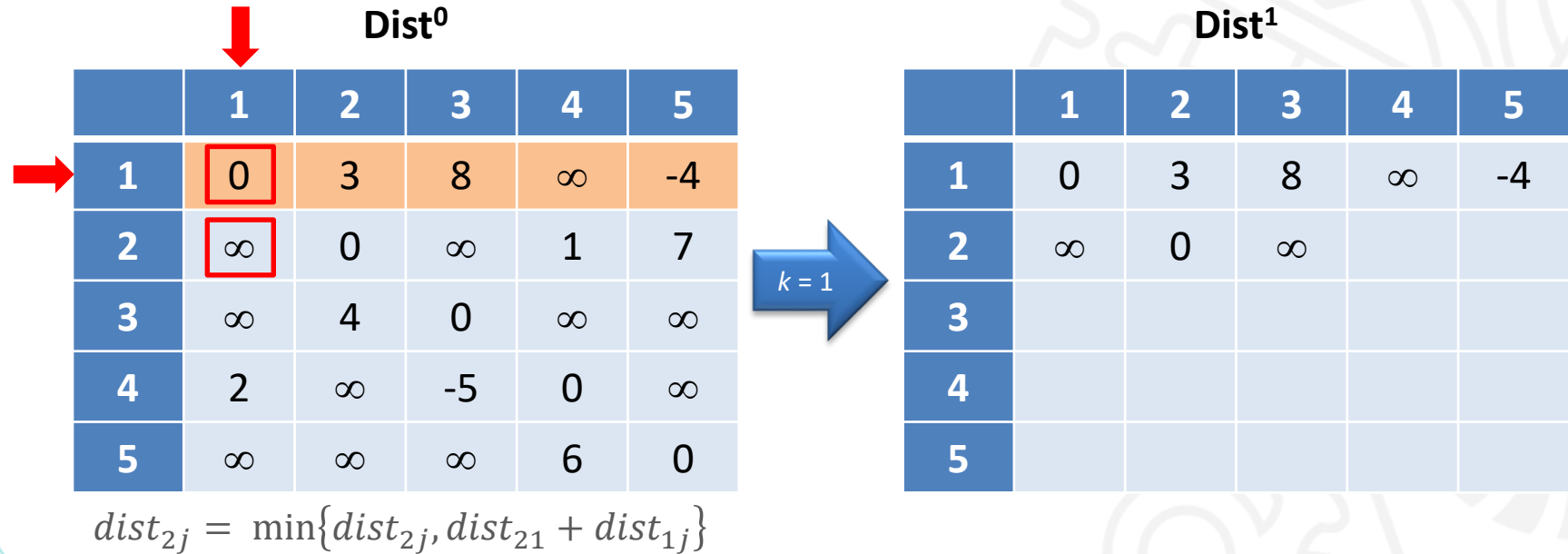
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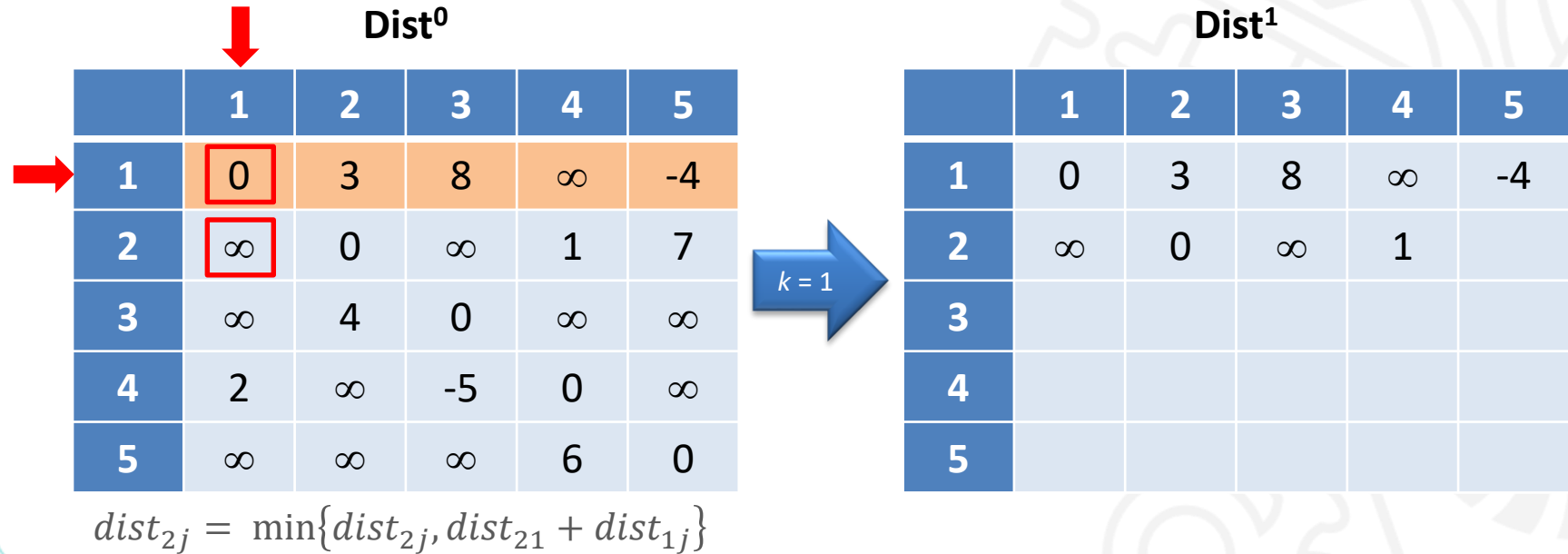
Método de Floyd-Warshall – Exemplo



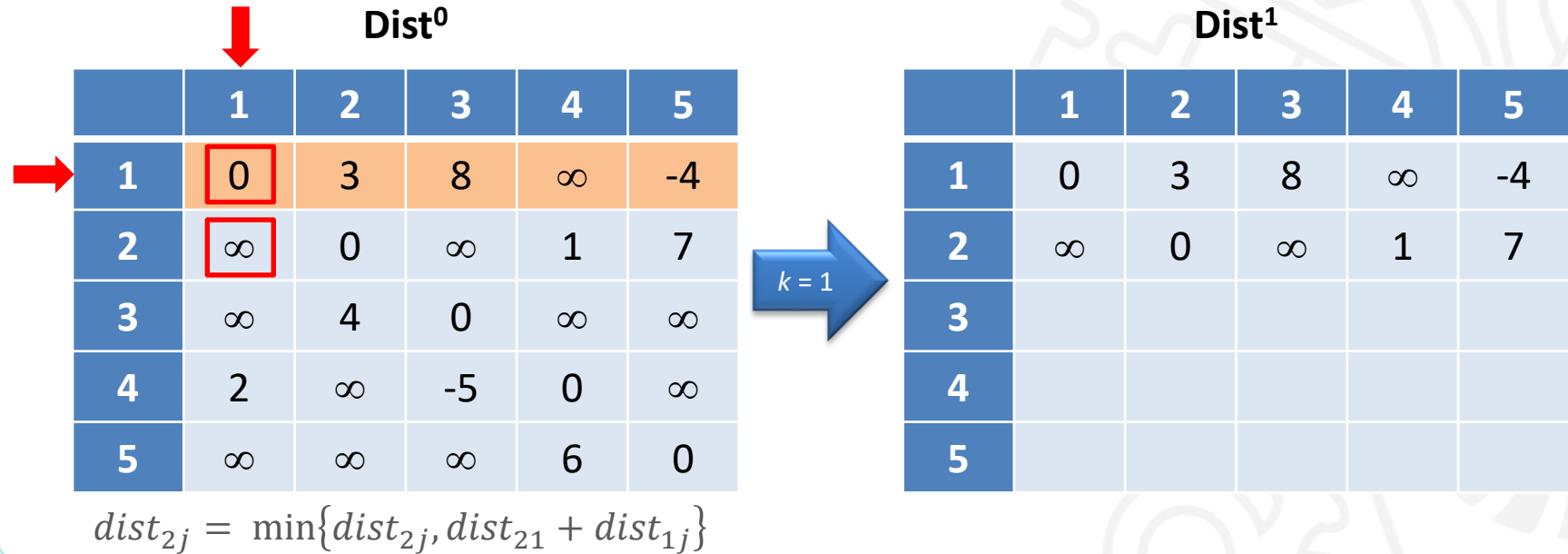
Método de Floyd-Warshall – Exemplo



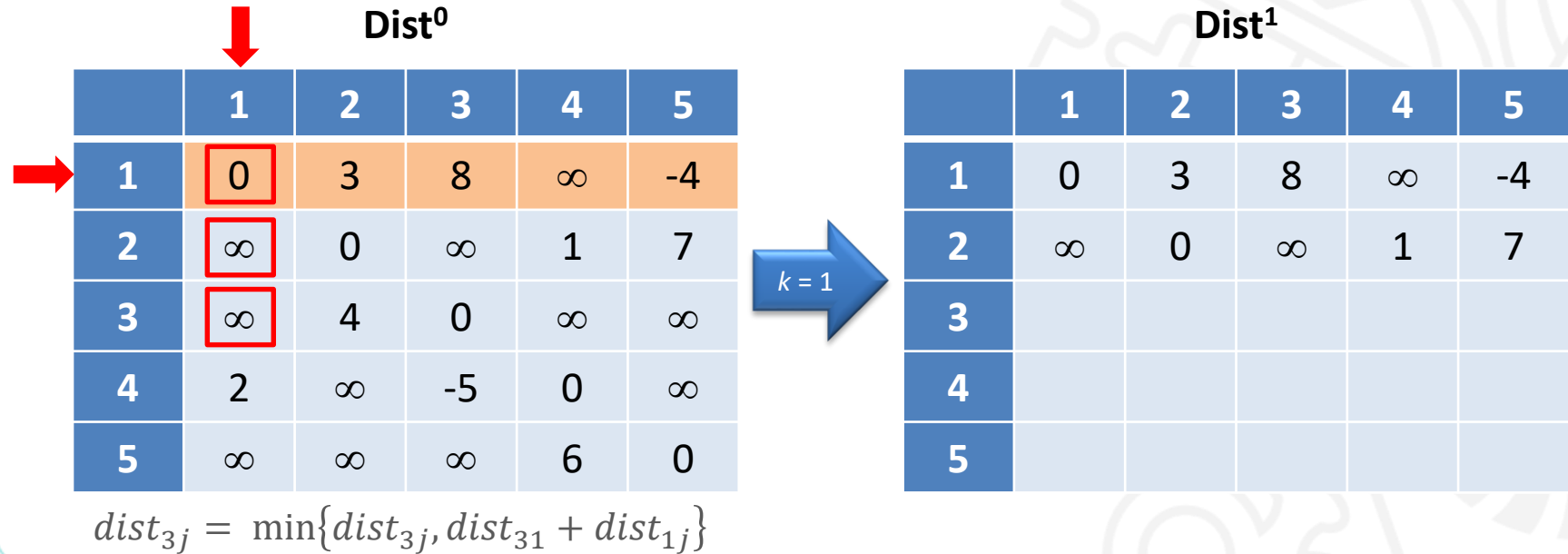
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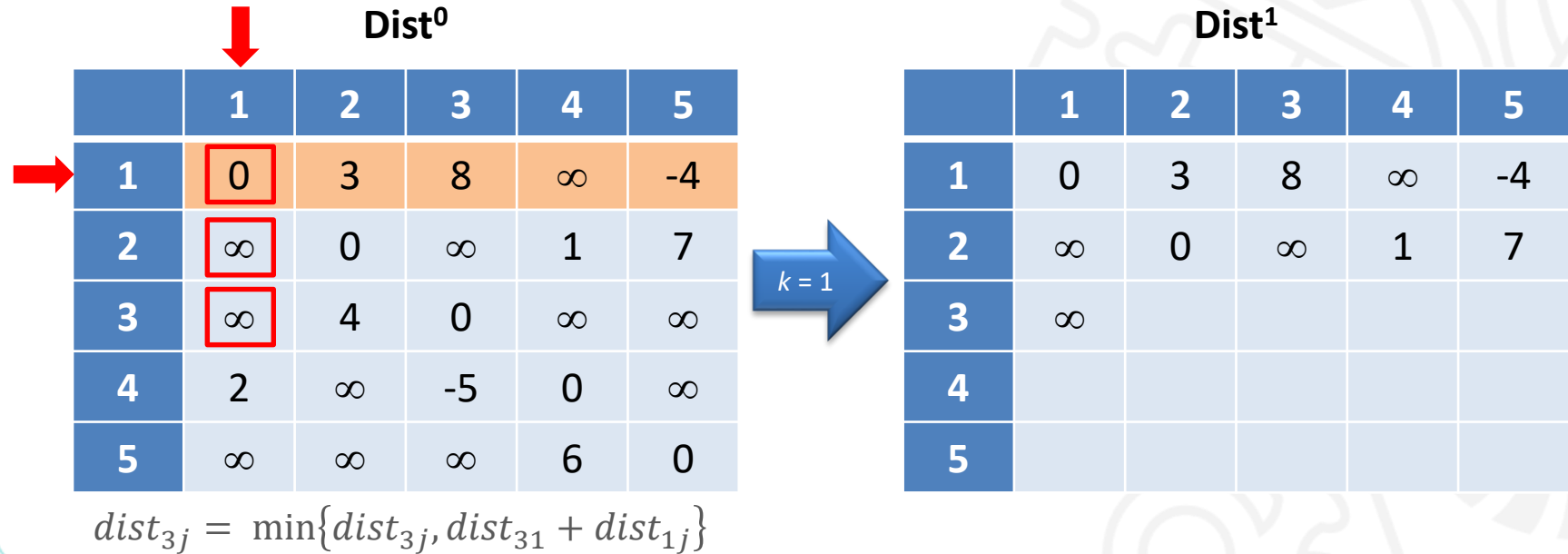
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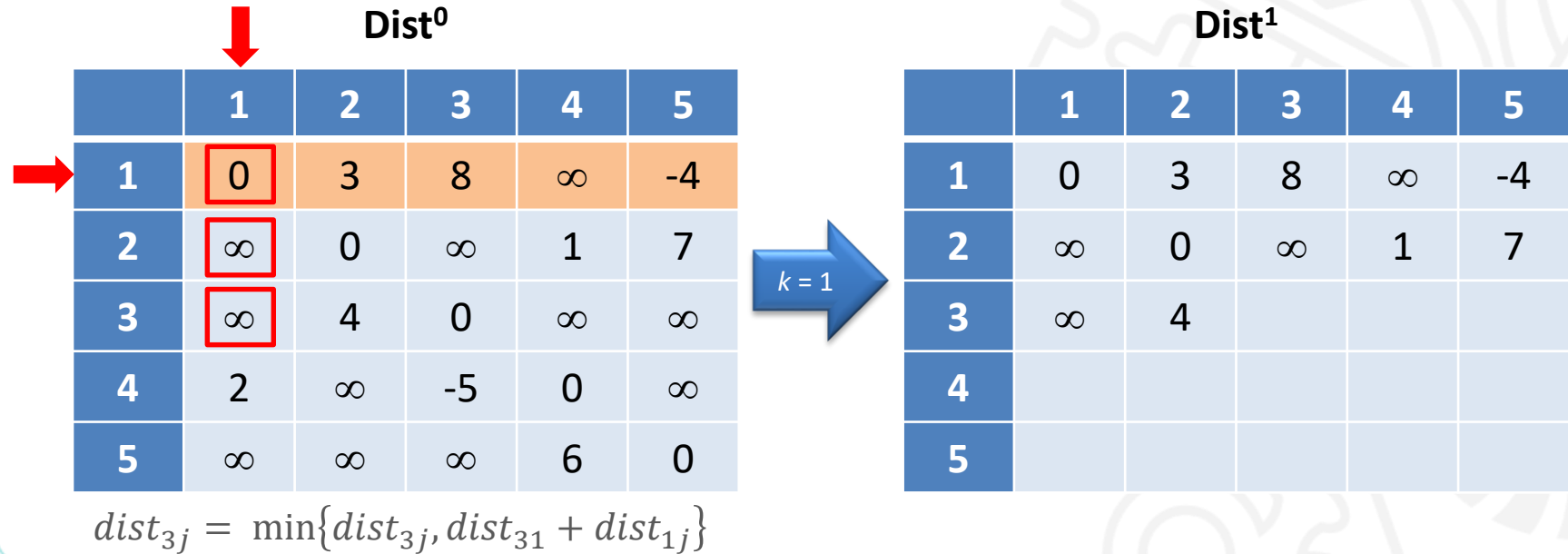
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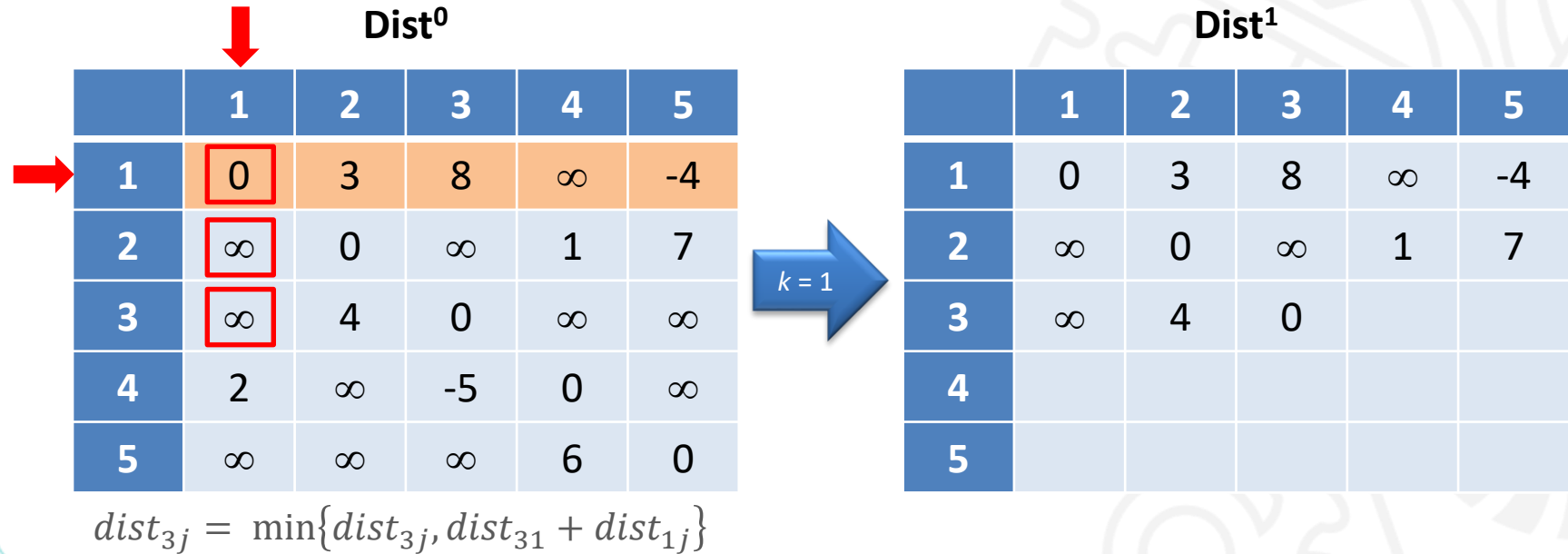
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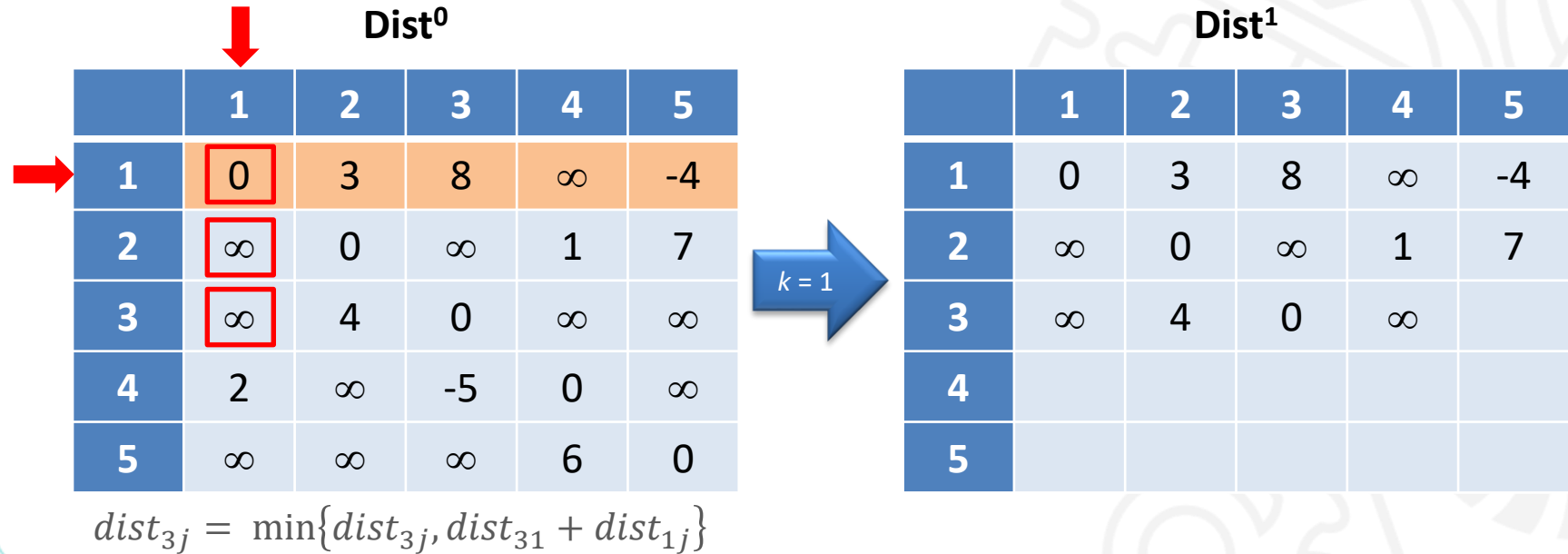
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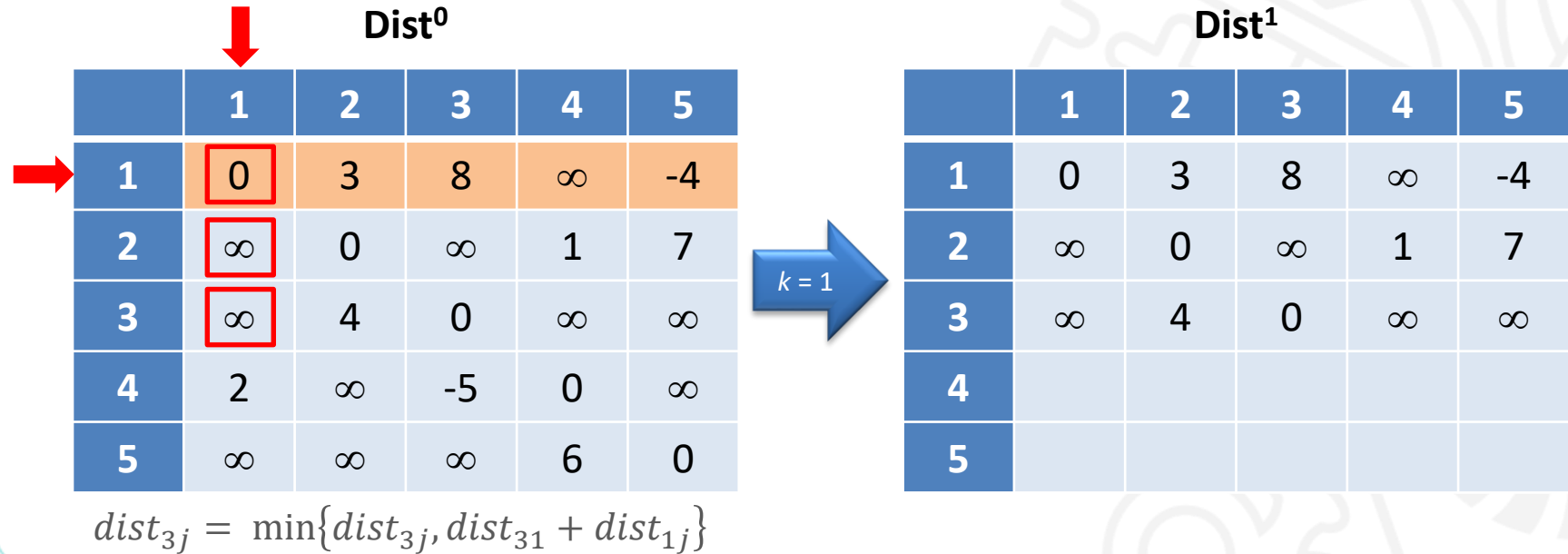
Método de Floyd-Warshall – Exemplo



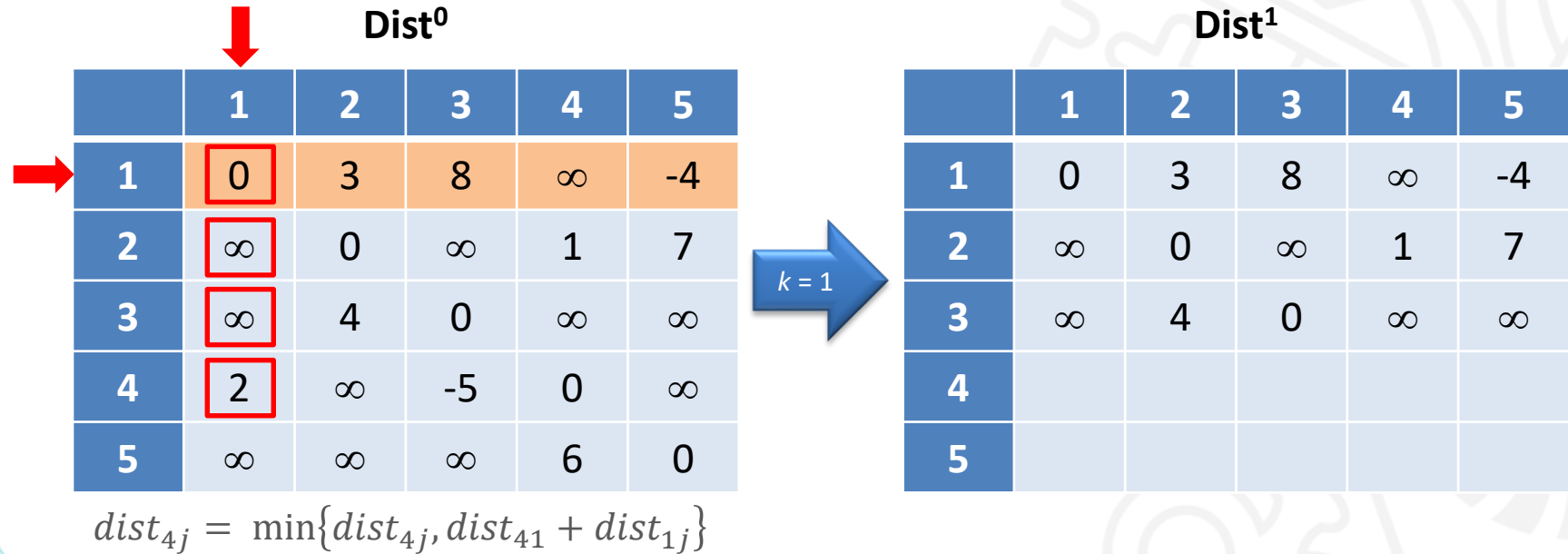
Método de Floyd-Warshall – Exemplo



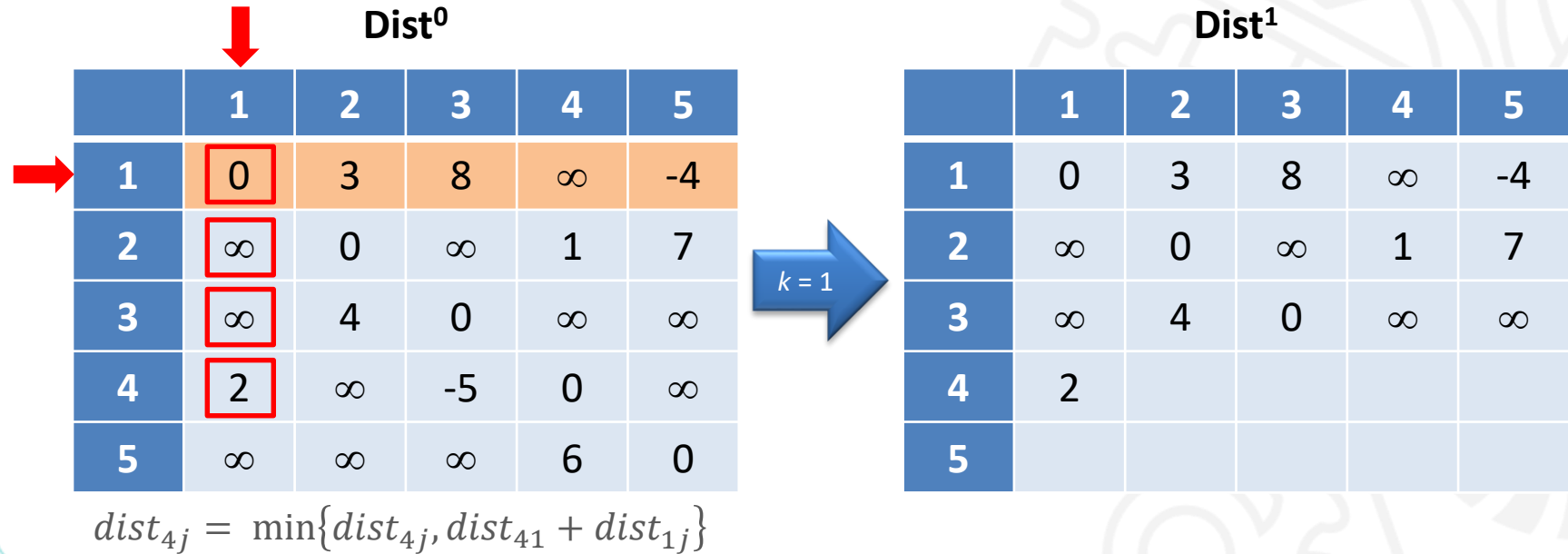
Método de Floyd-Warshall – Exemplo



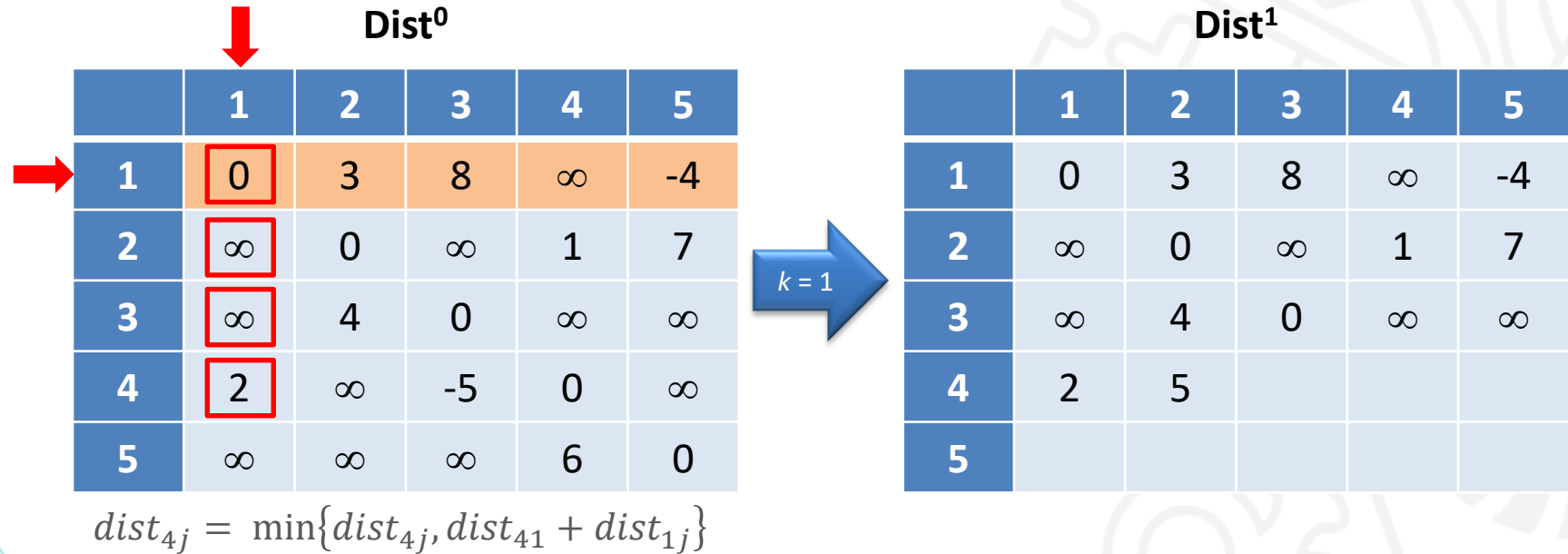
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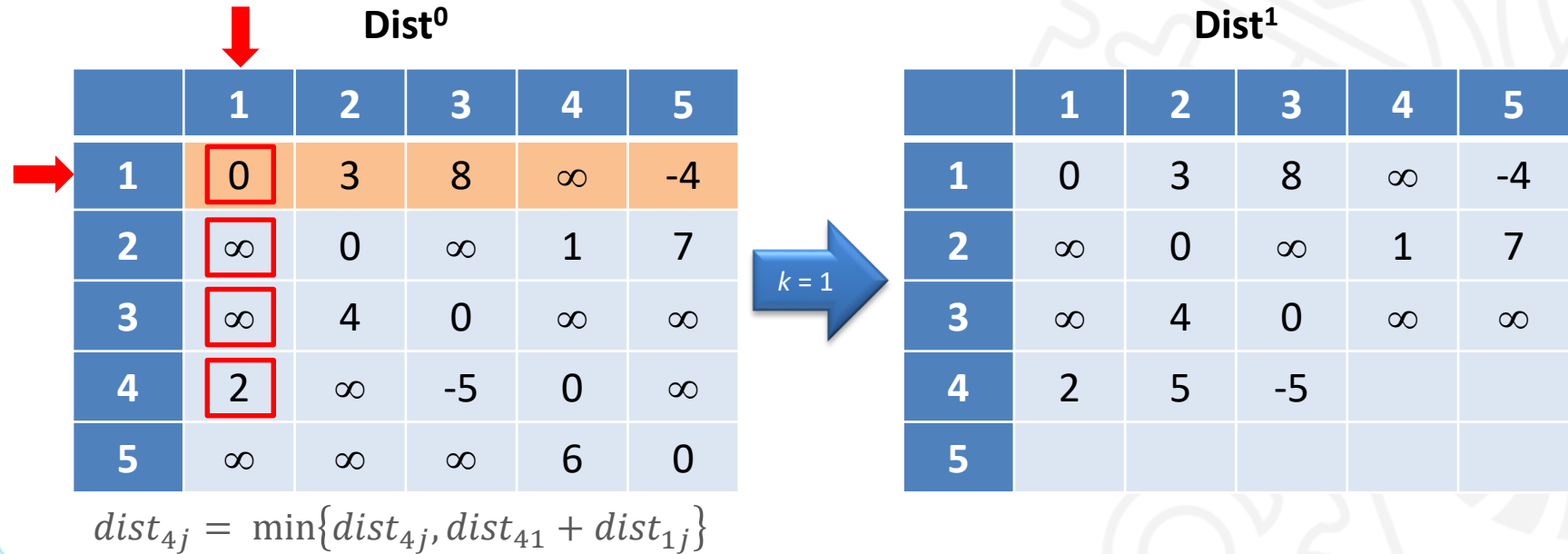
Método de Floyd-Warshall – Exemplo



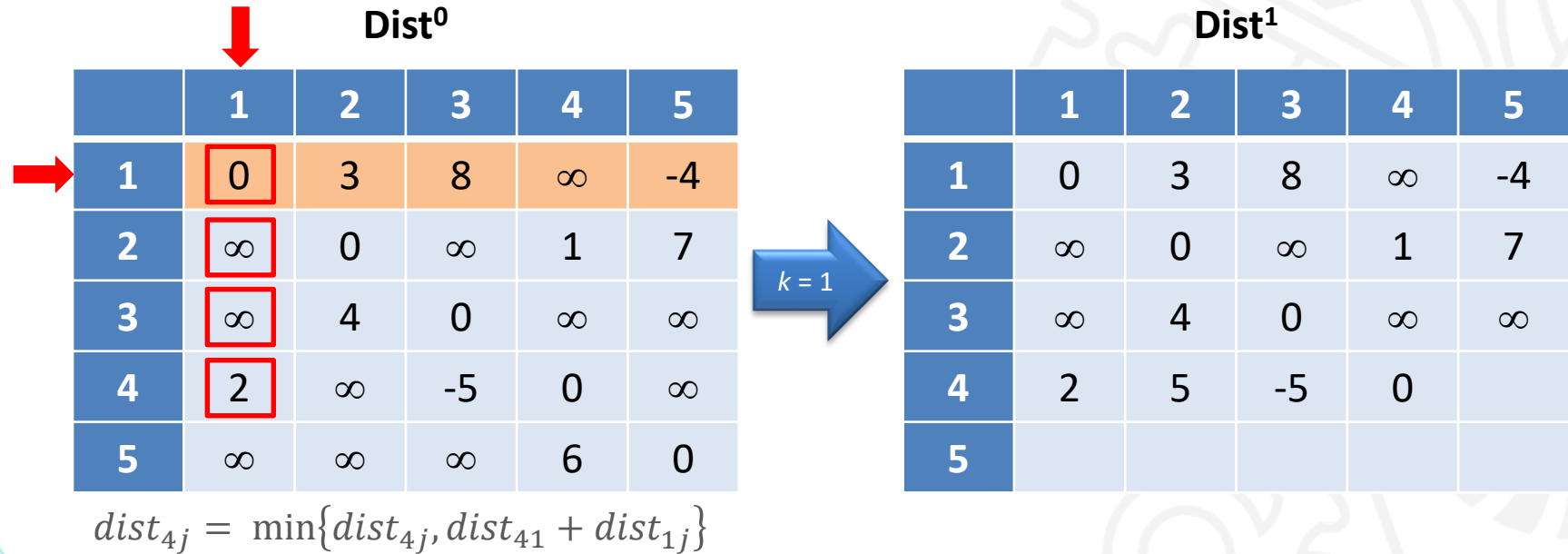
Método de Floyd-Warshall – Exemplo



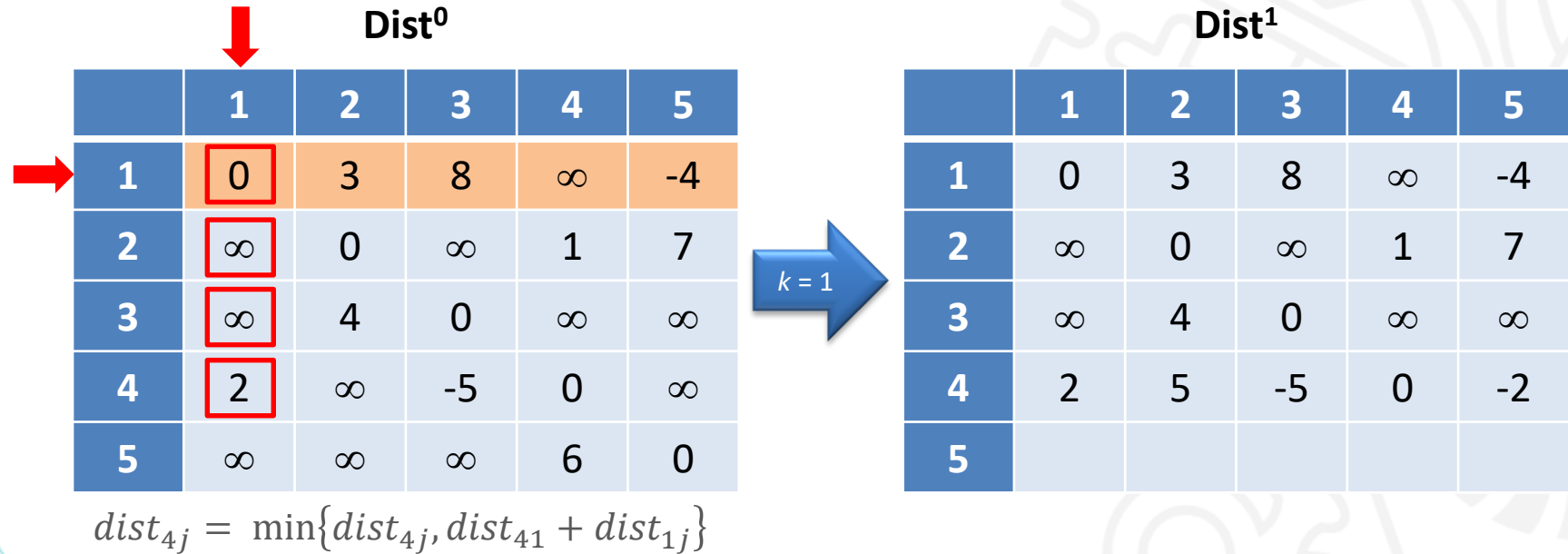
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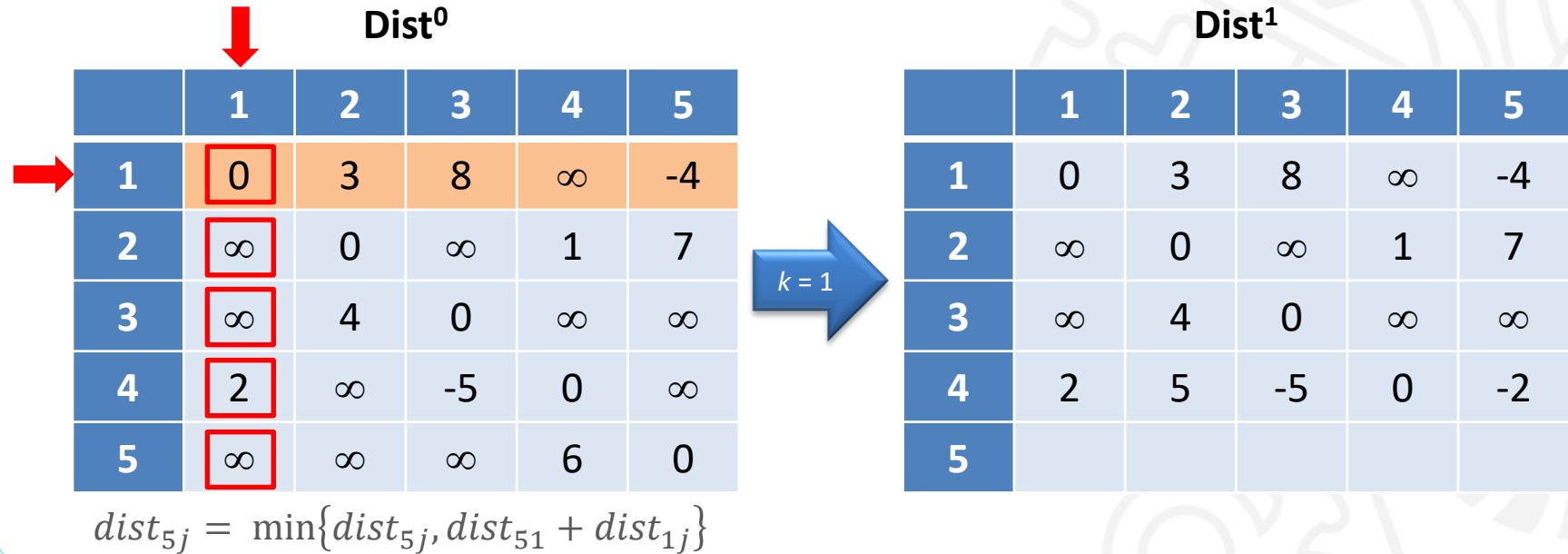
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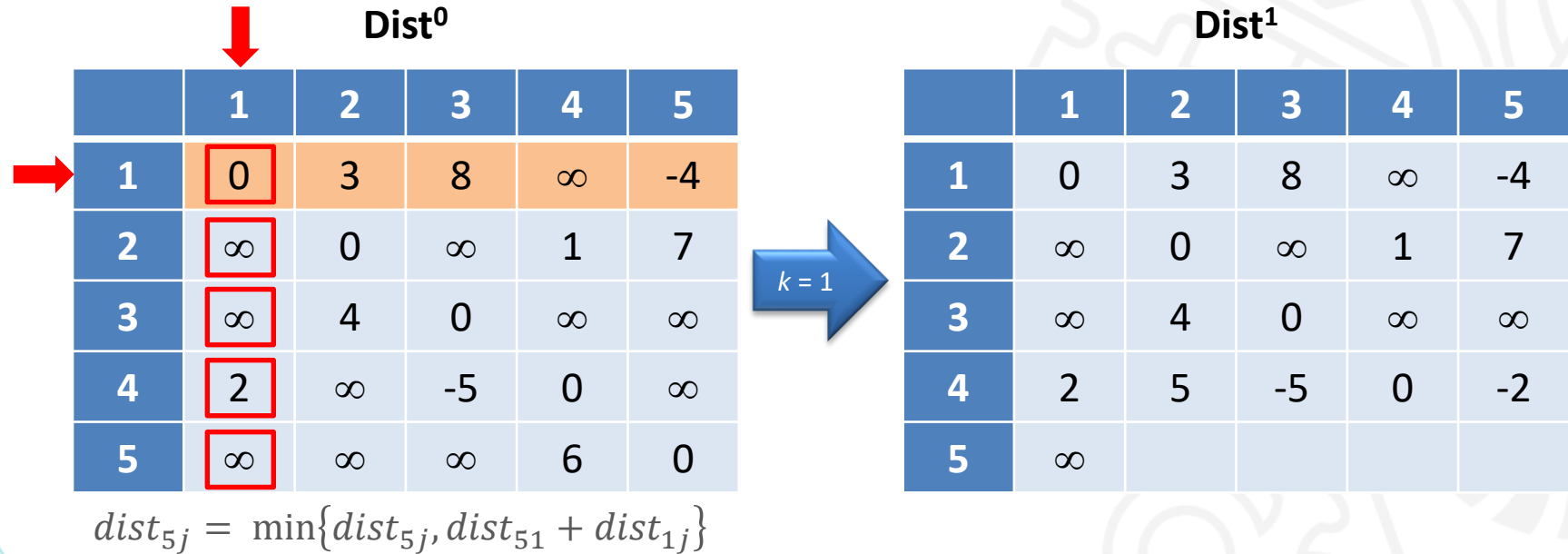
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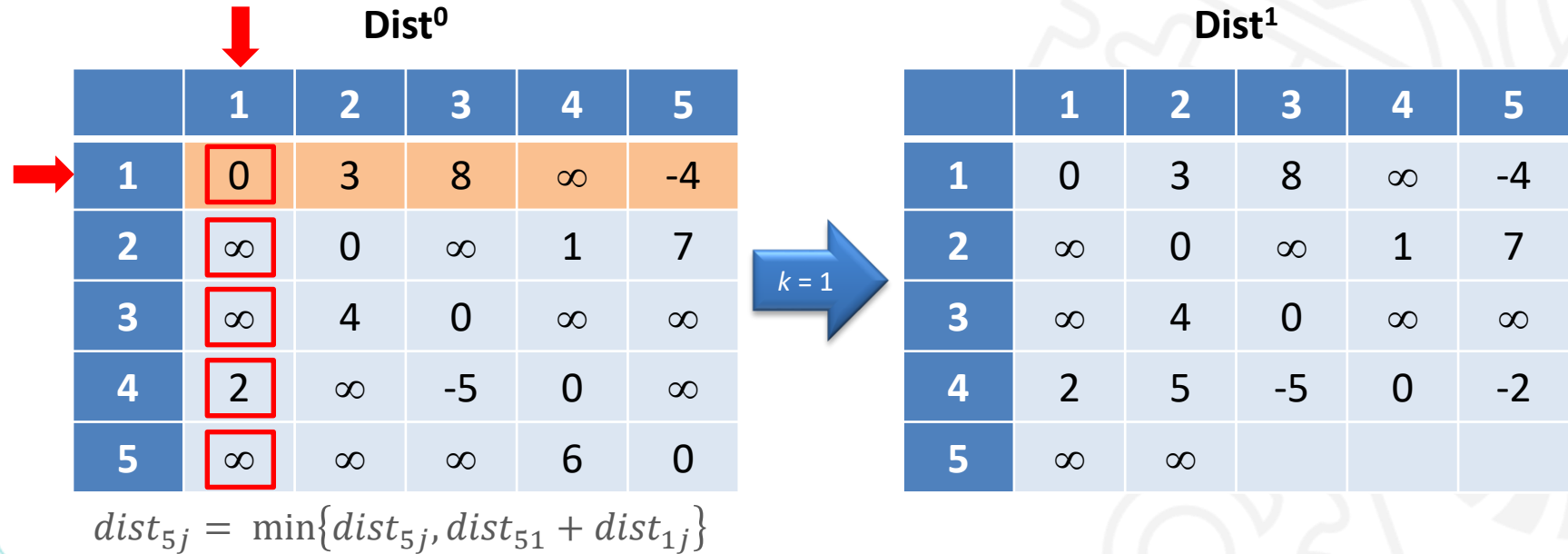
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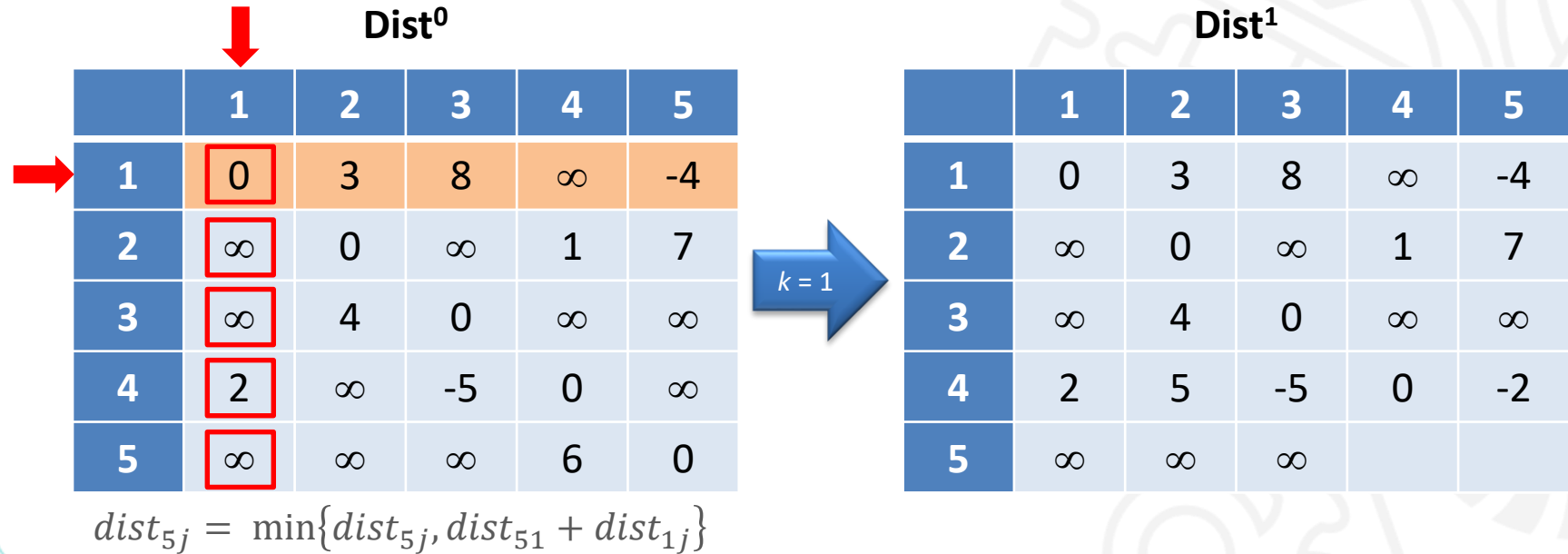
Método de Floyd-Warshall – Exemplo



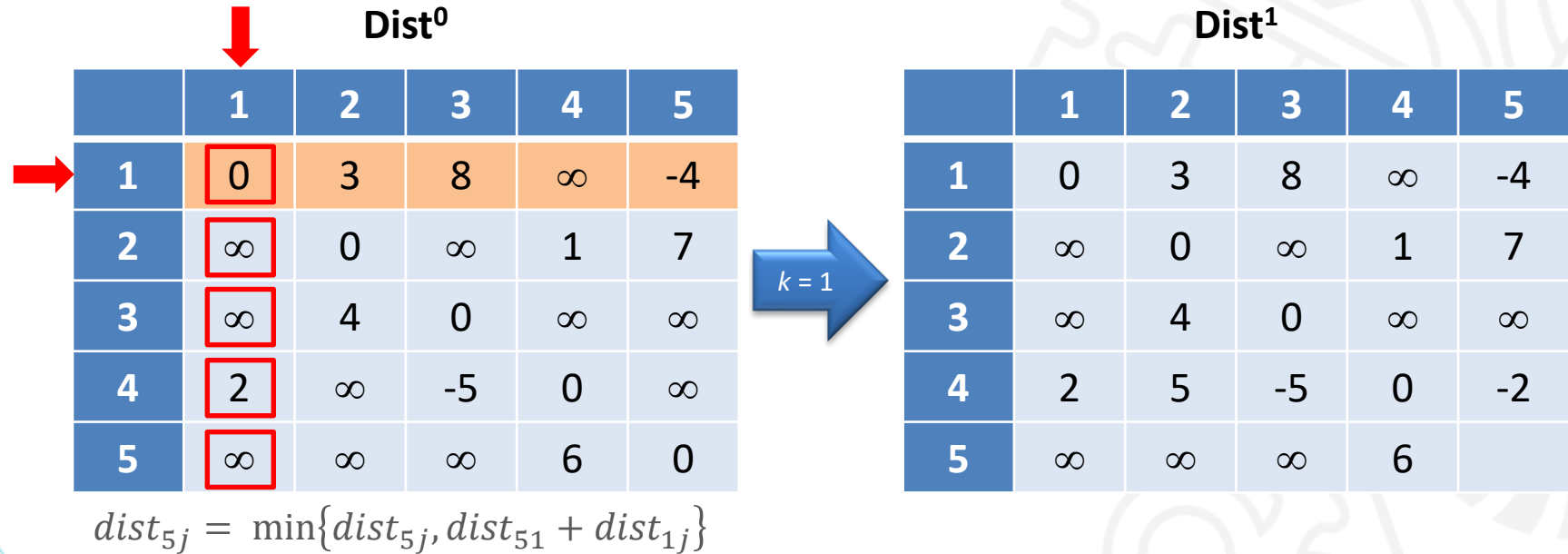
Método de Floyd-Warshall – Exemplo



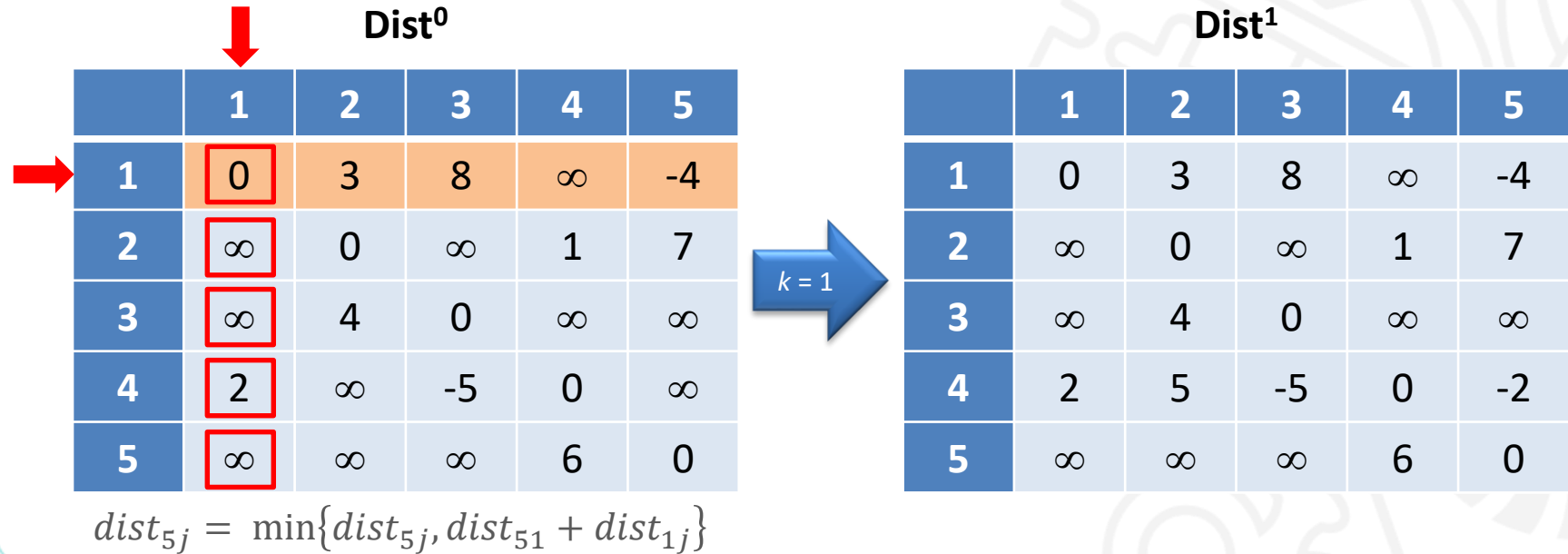
Método de Floyd-Warshall – Exemplo



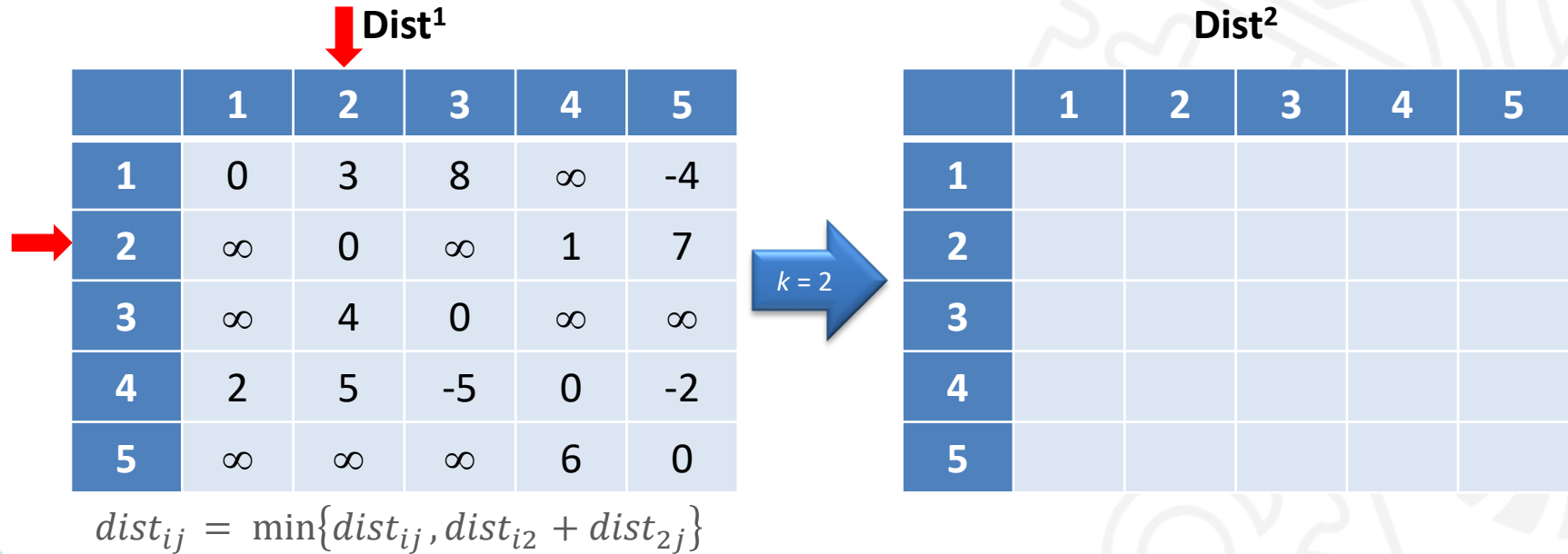
Método de Floyd-Warshall – Exemplo



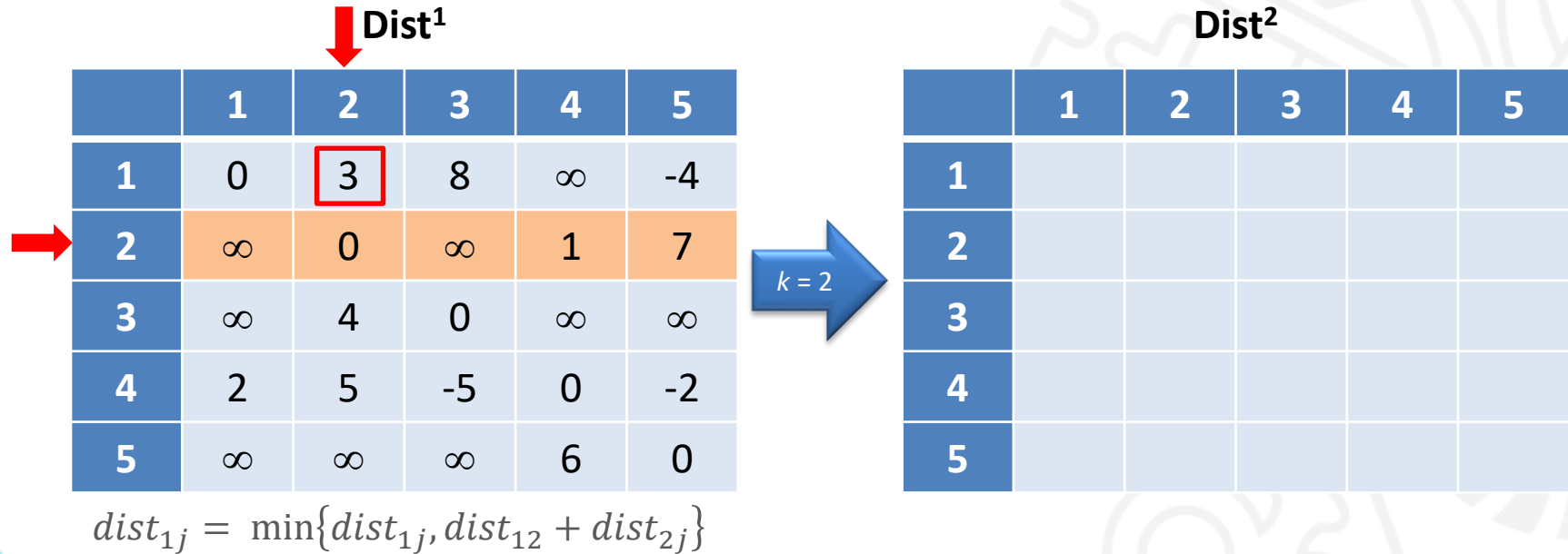
Método de Floyd-Warshall – Exemplo



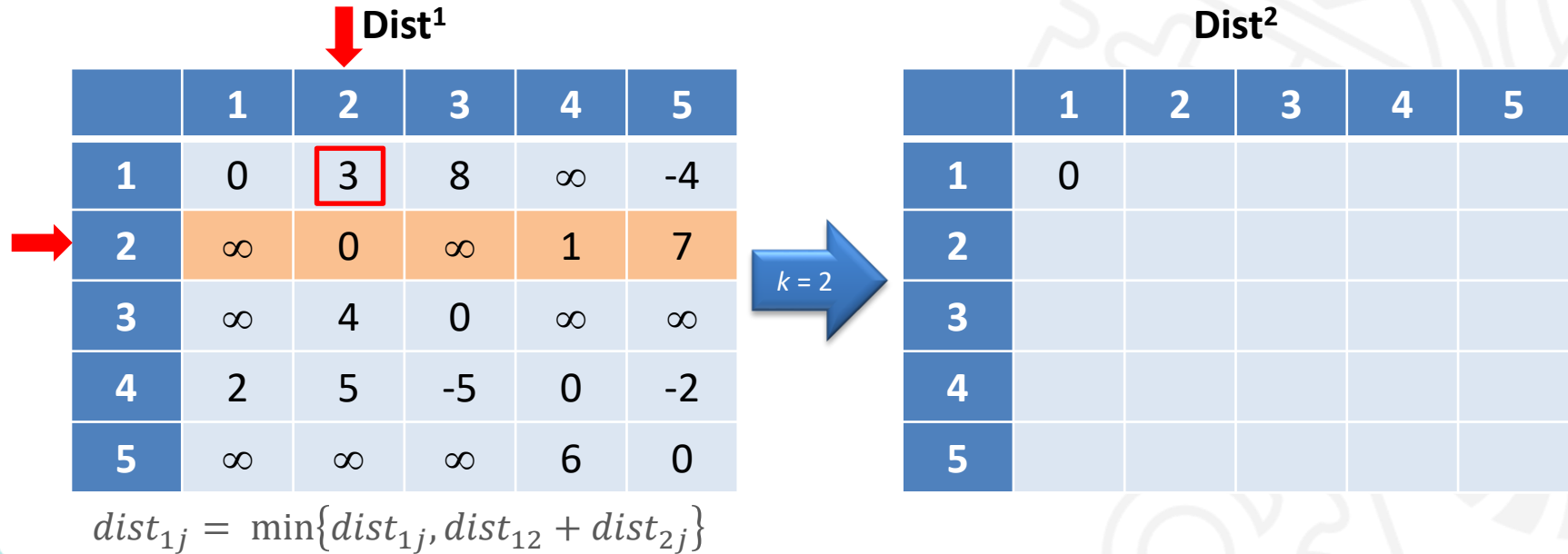
Método de Floyd-Warshall – Exemplo



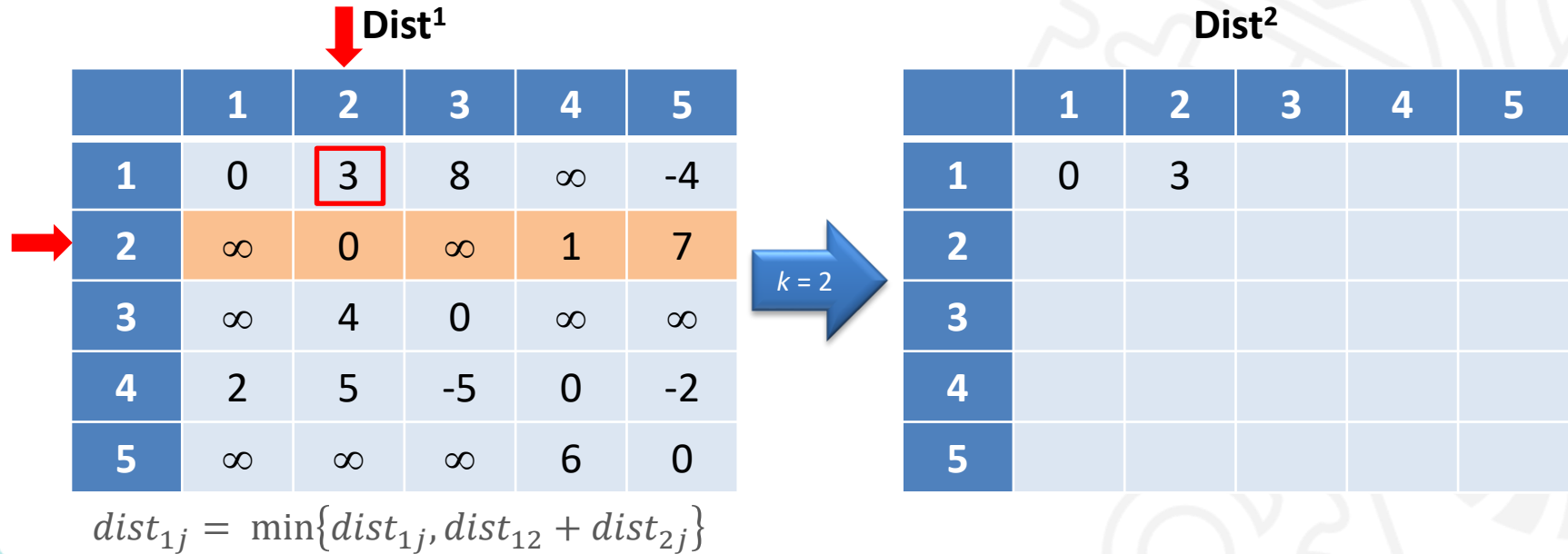
Método de Floyd-Warshall – Exemplo



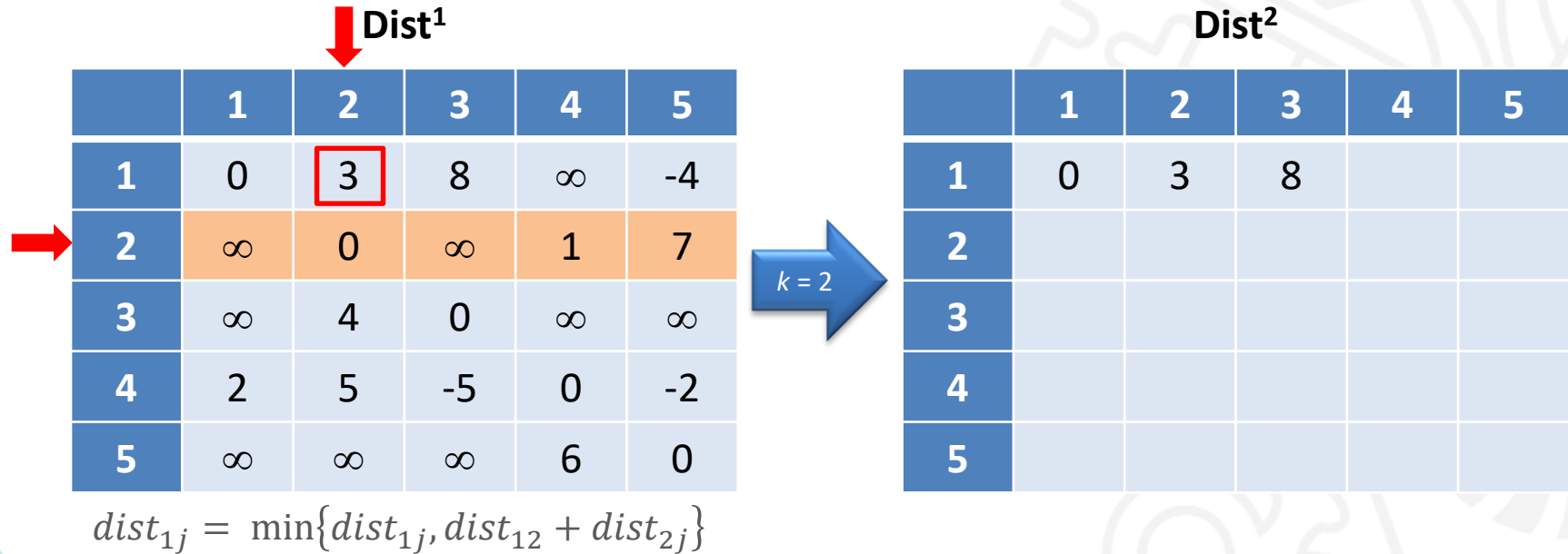
Método de Floyd-Warshall – Exemplo



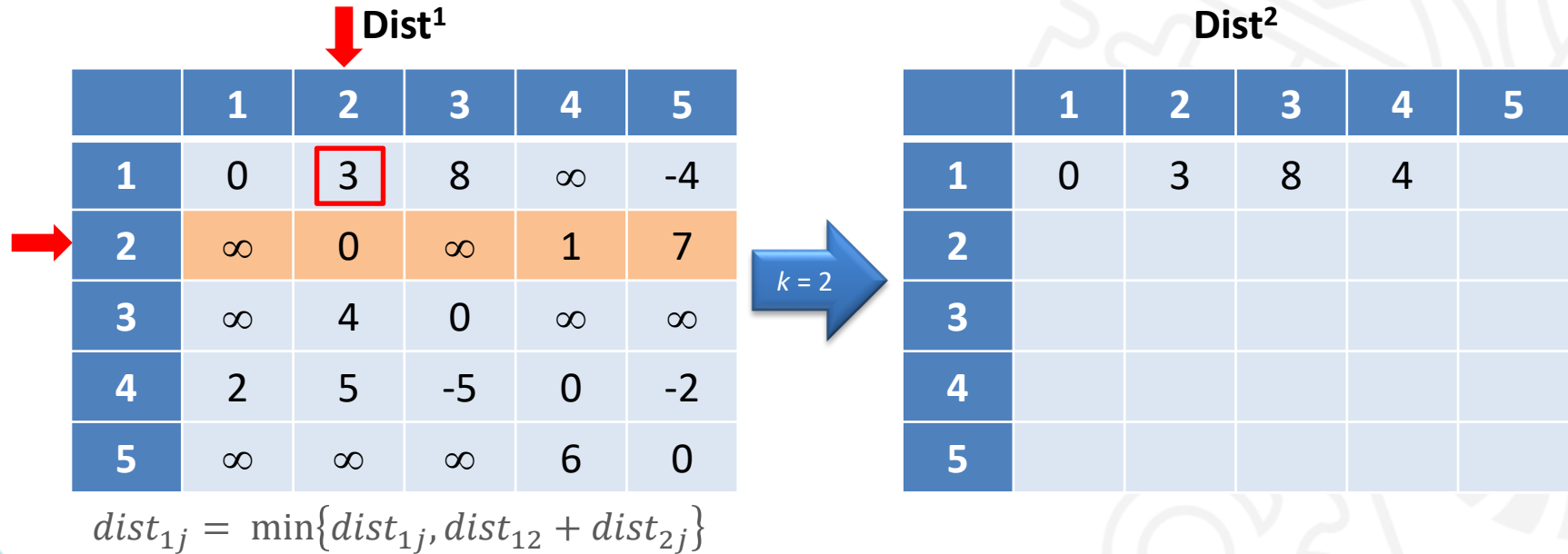
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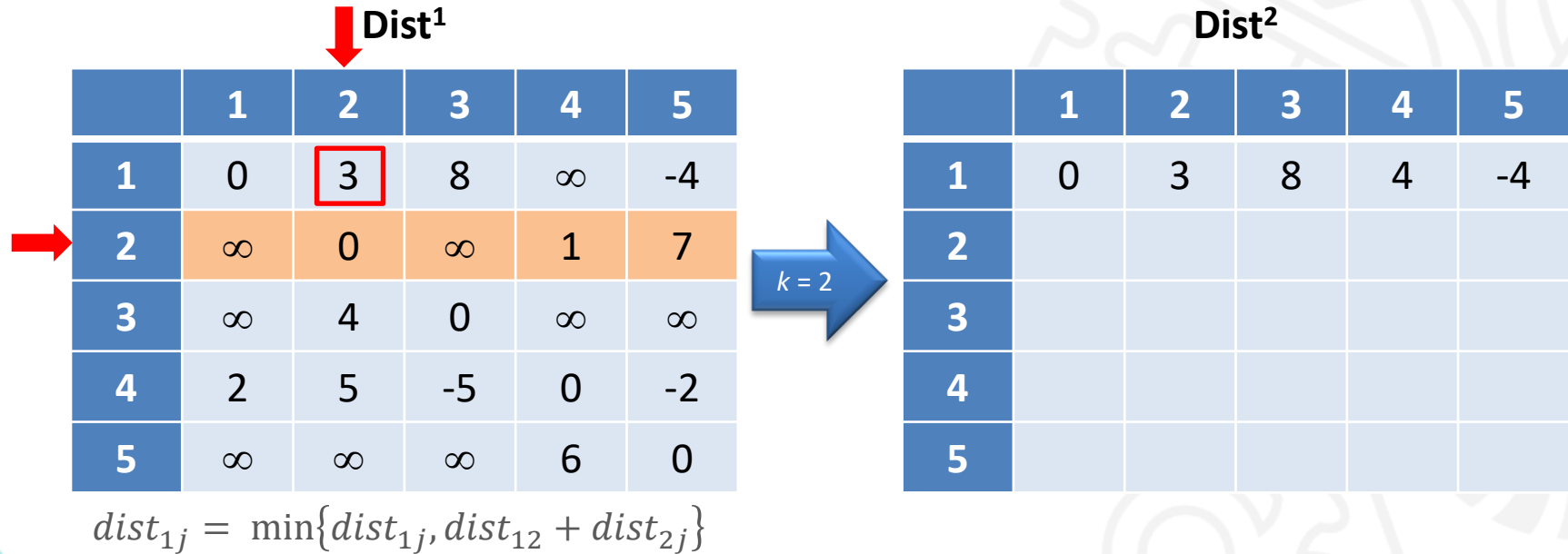
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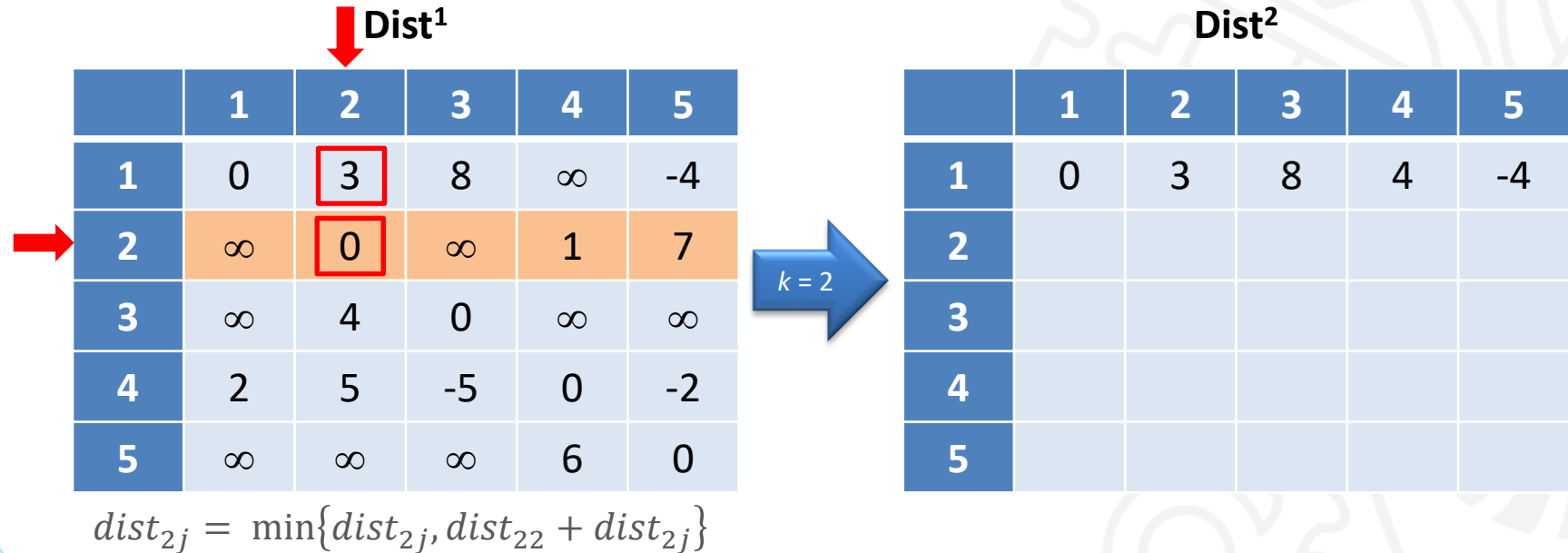
Método de Floyd-Warshall – Exemplo



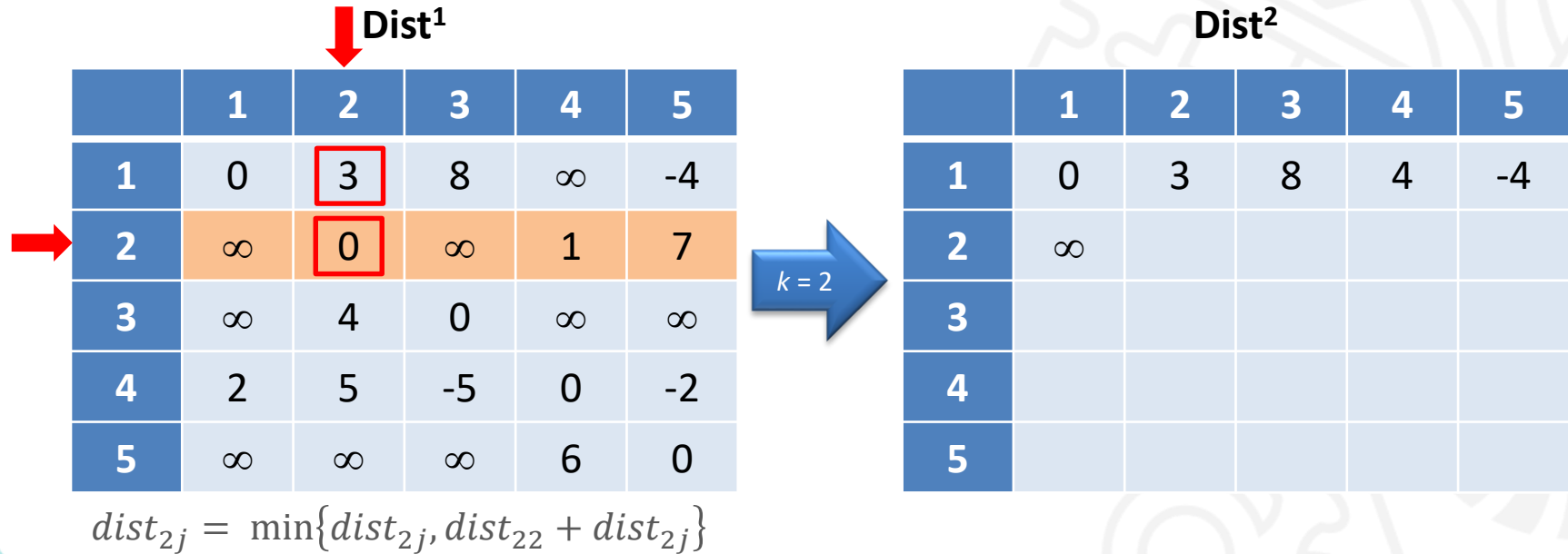
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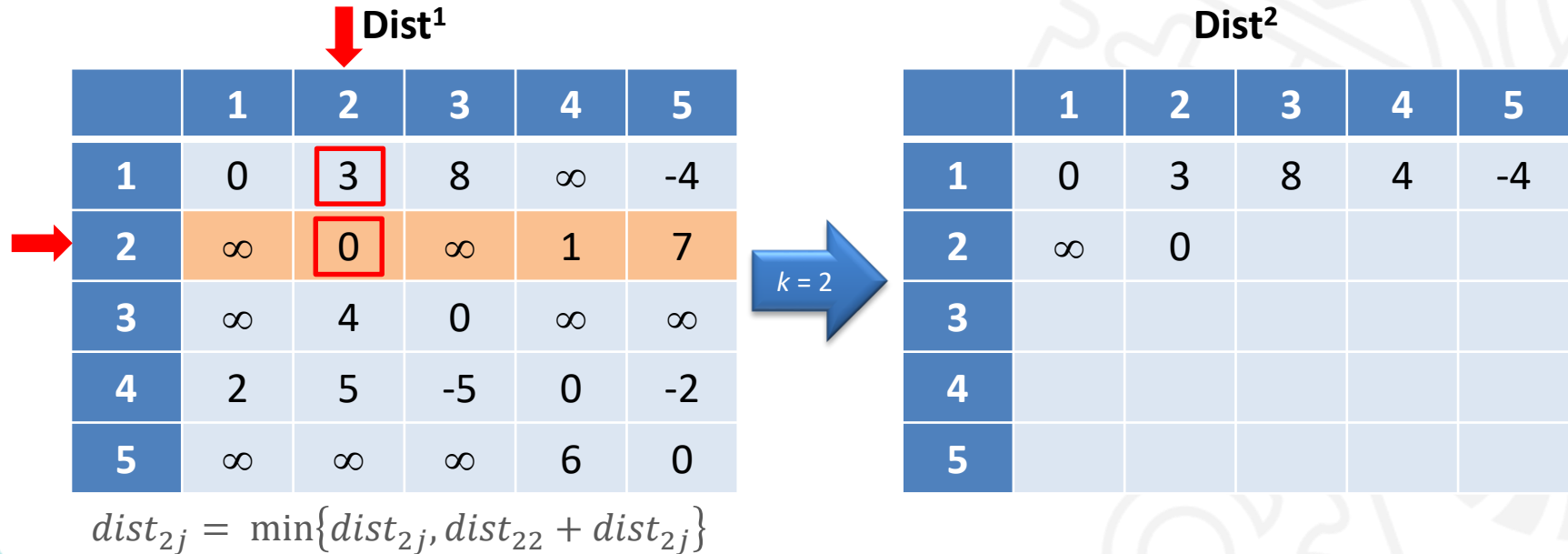
Método de Floyd-Warshall – Exemplo



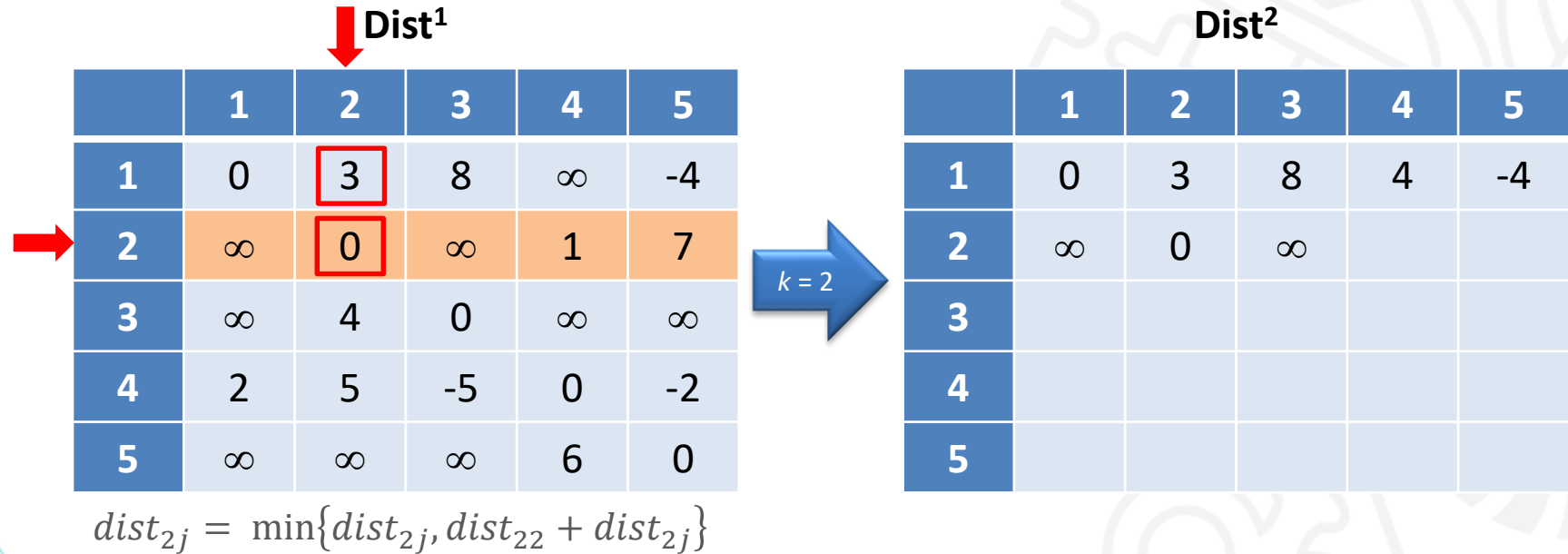
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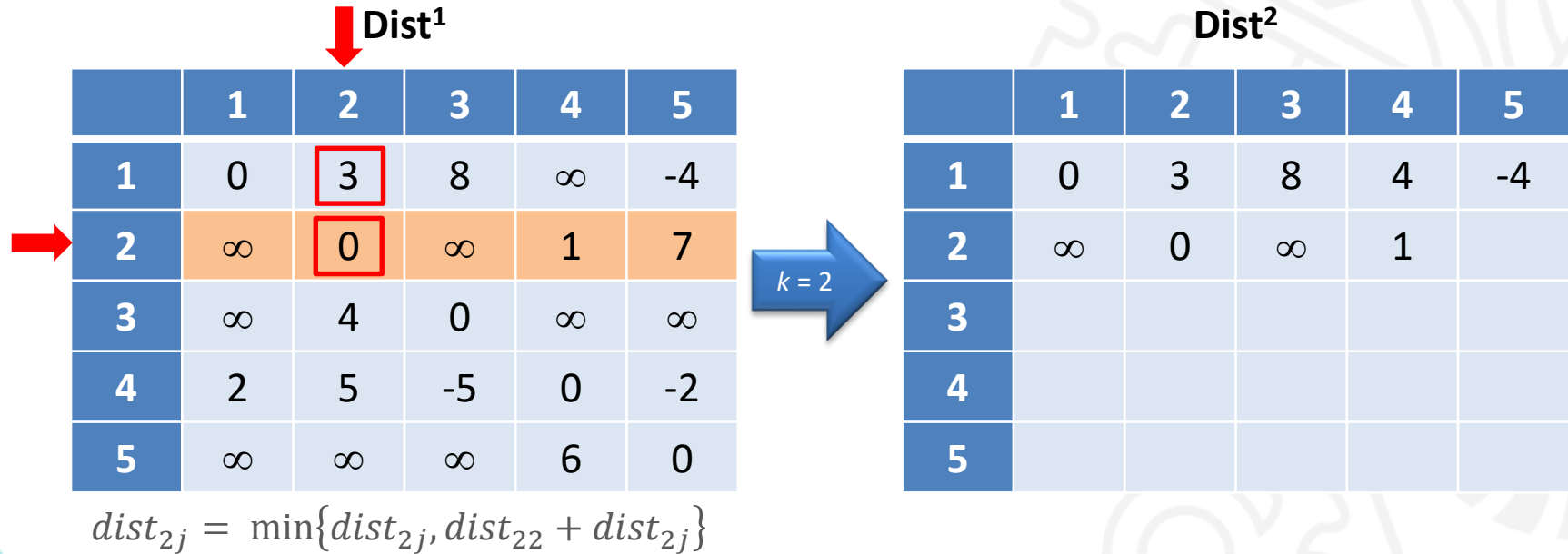
Método de Floyd-Warshall – Exemplo



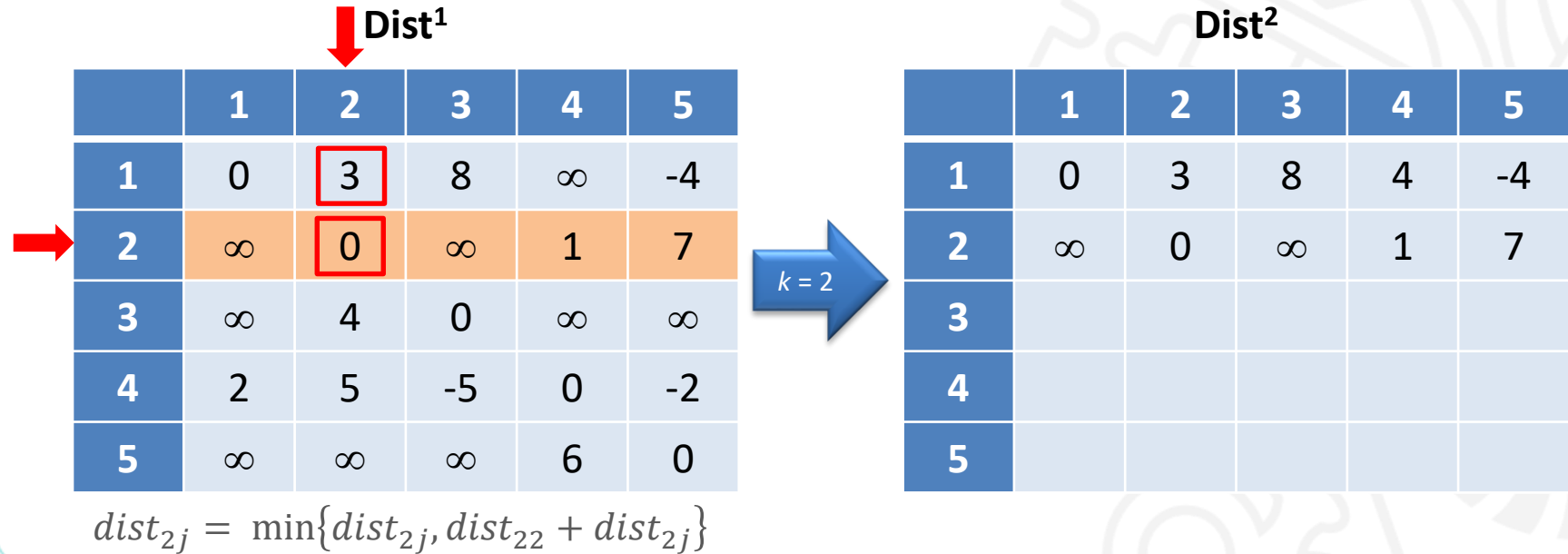
Método de Floyd-Warshall – Exemplo



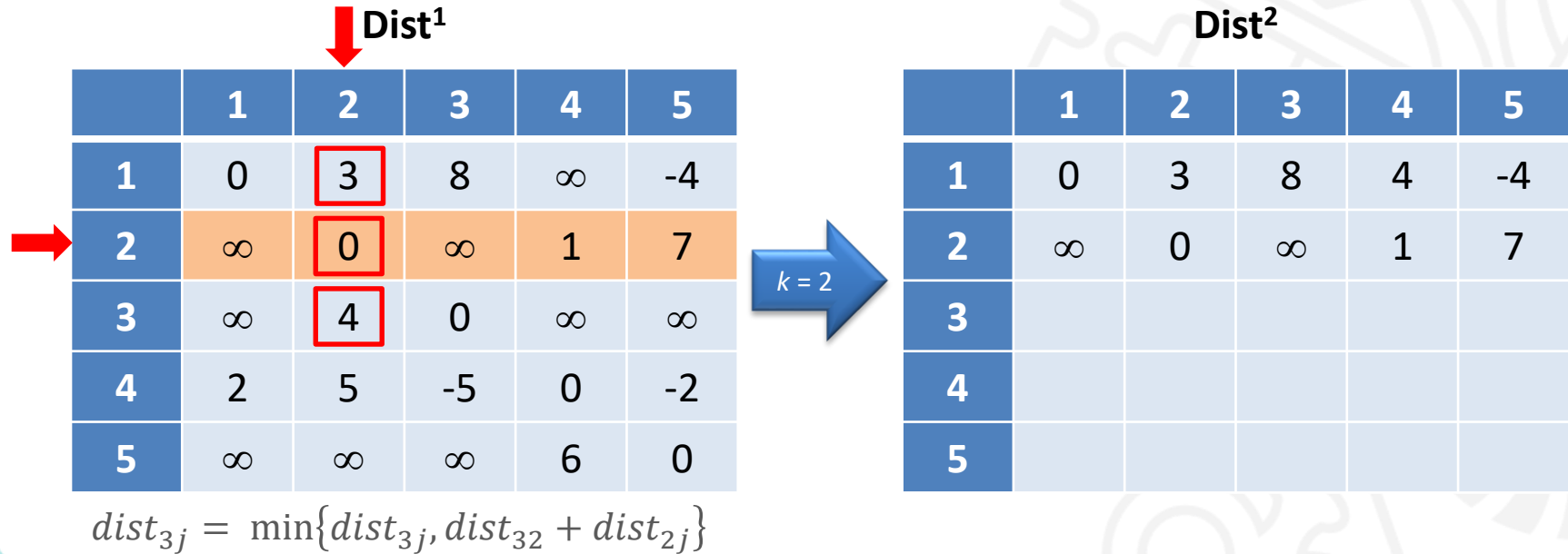
Método de Floyd-Warshall – Exemplo



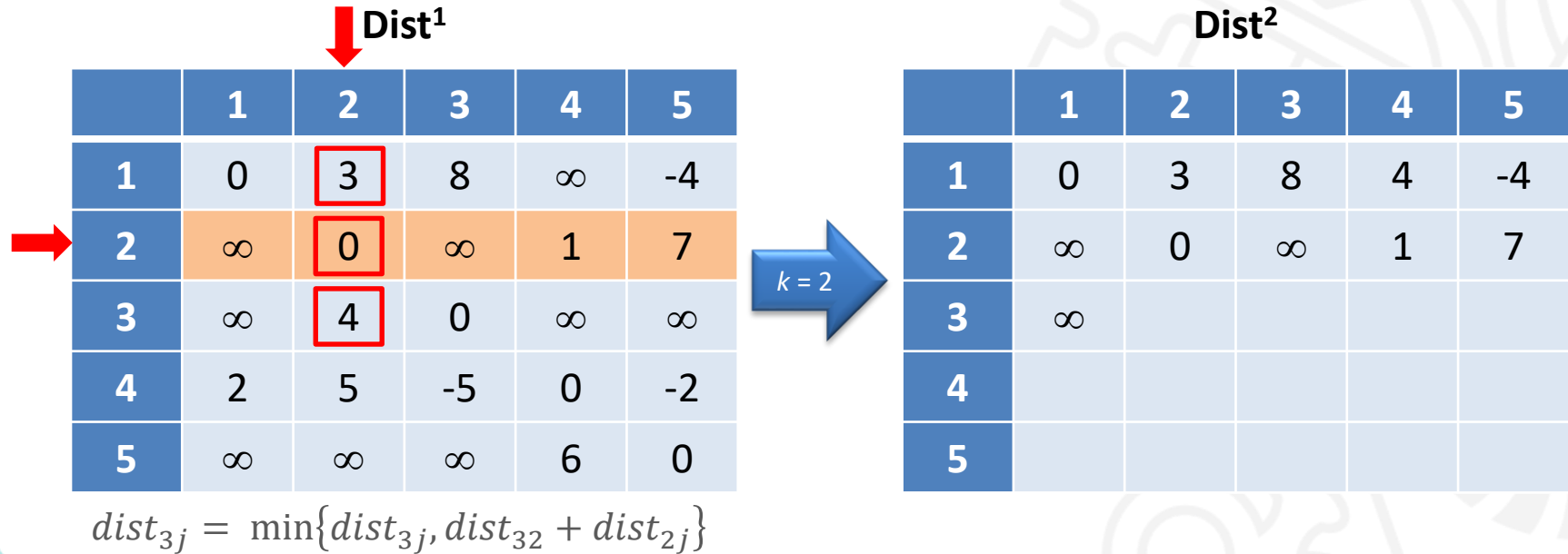
Método de Floyd-Warshall – Exemplo



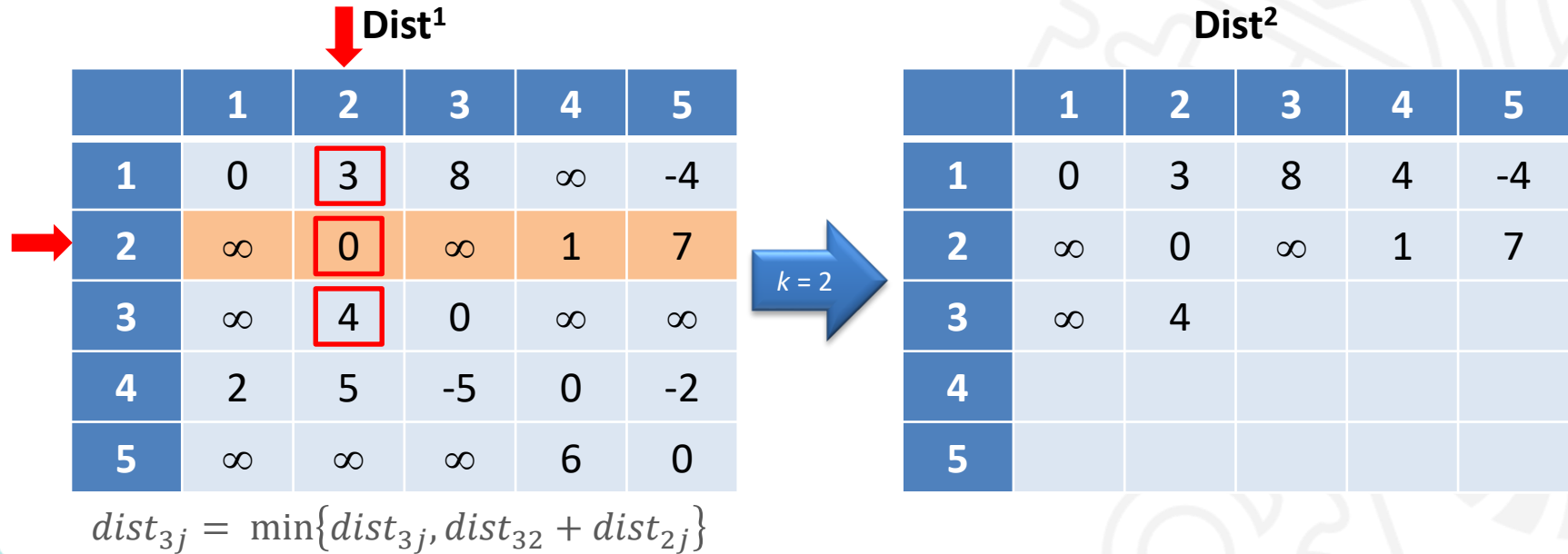
Método de Floyd-Warshall – Exemplo



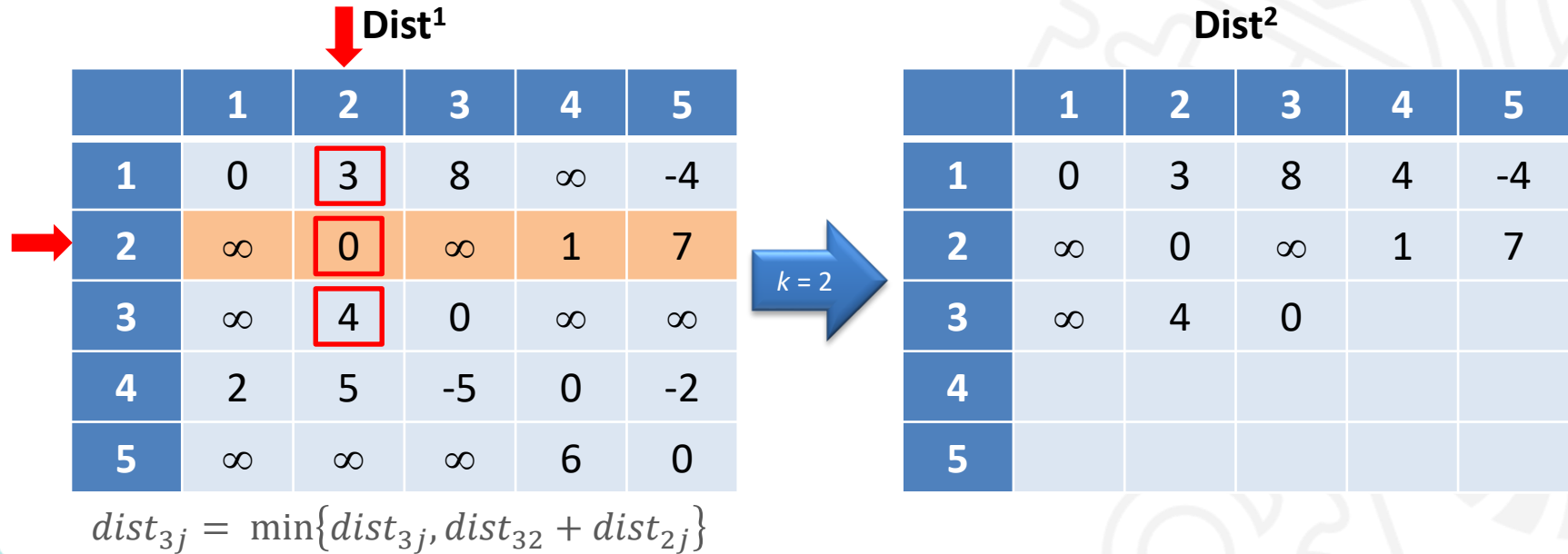
Método de Floyd-Warshall – Exemplo



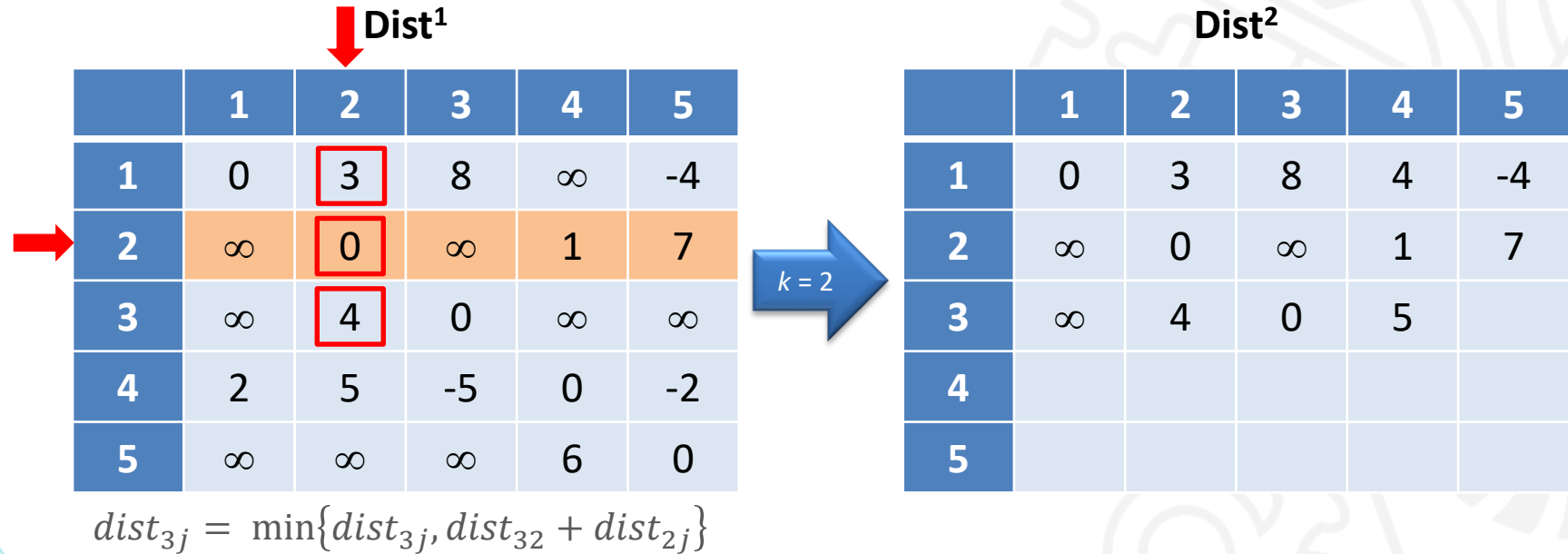
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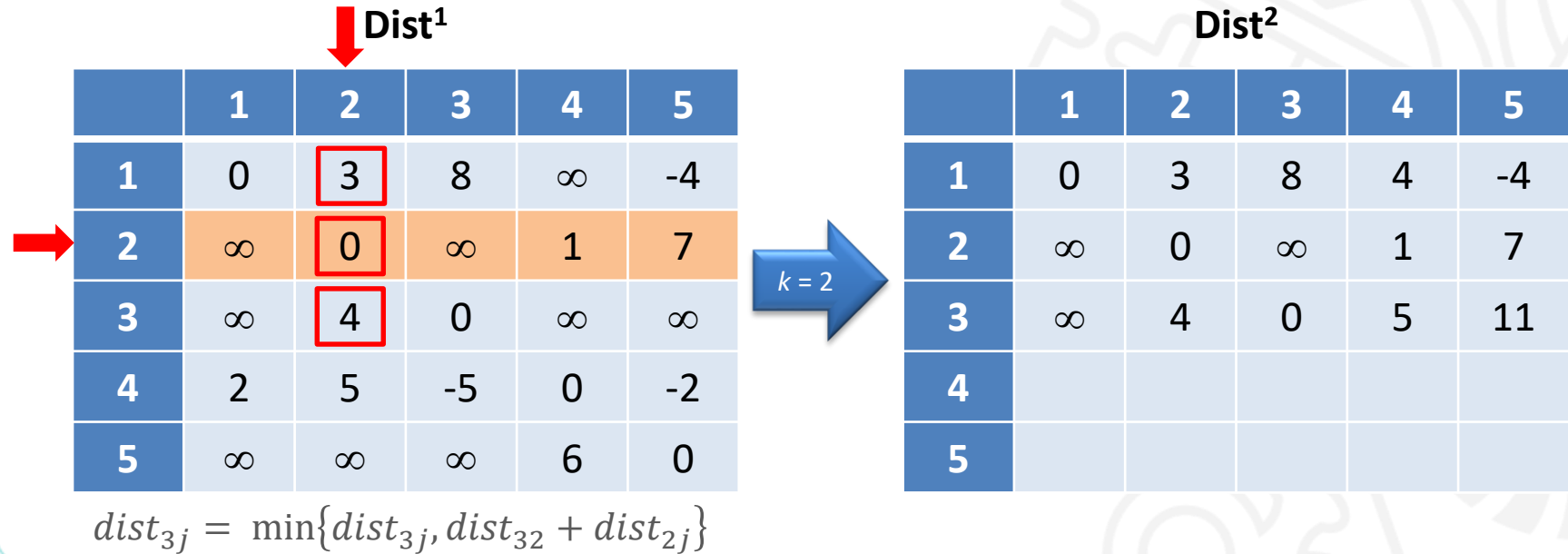
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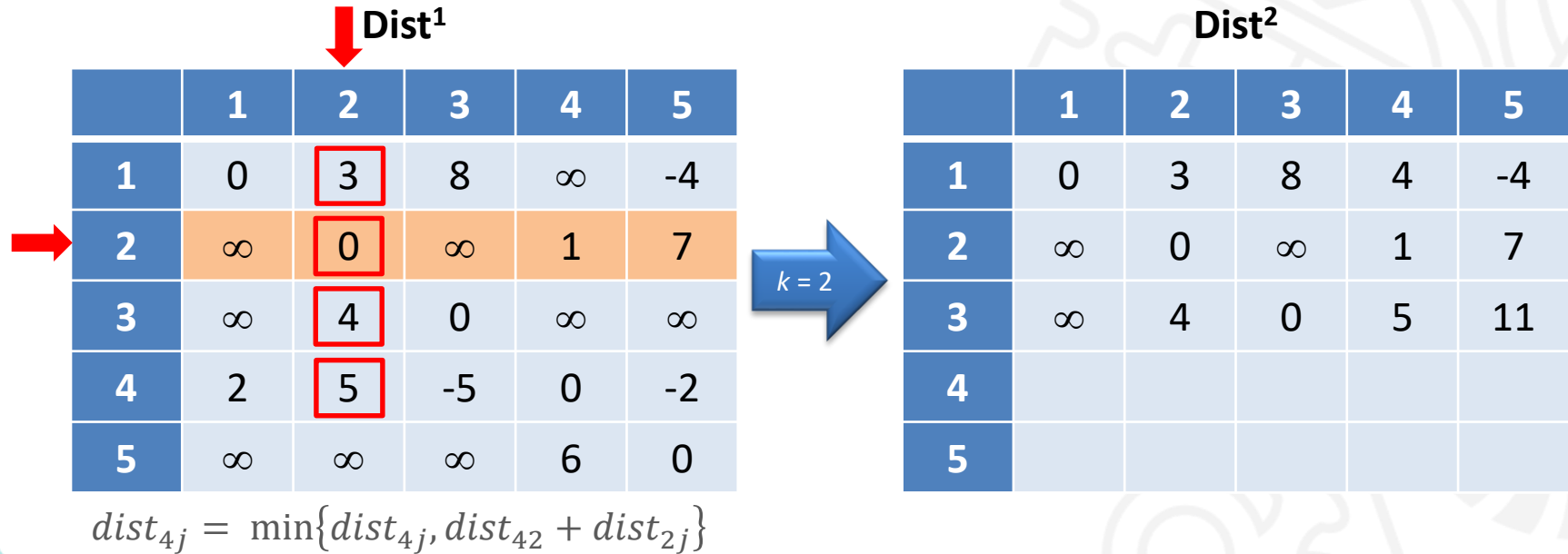
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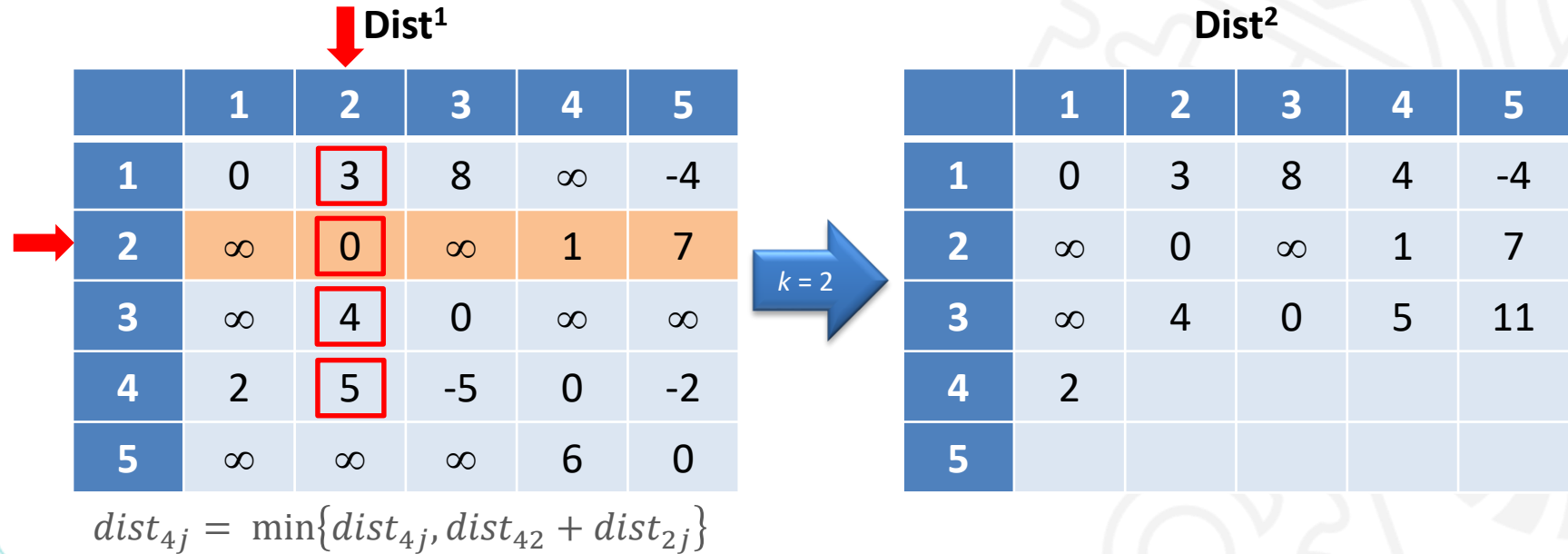
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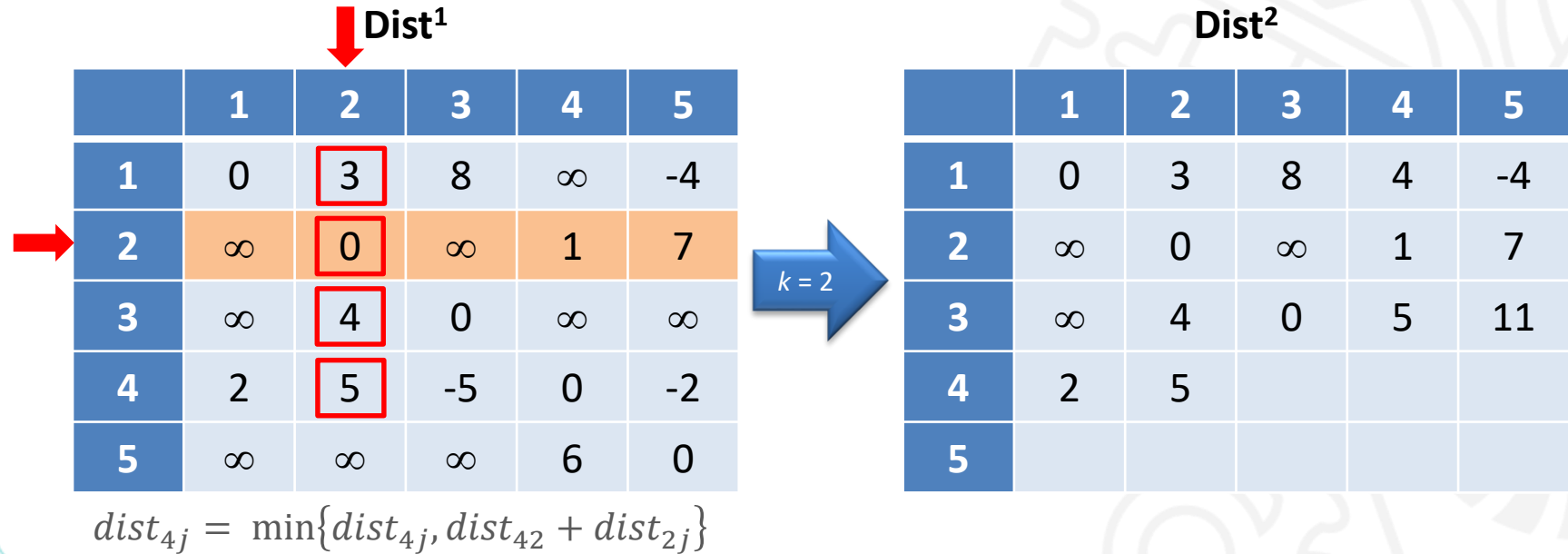
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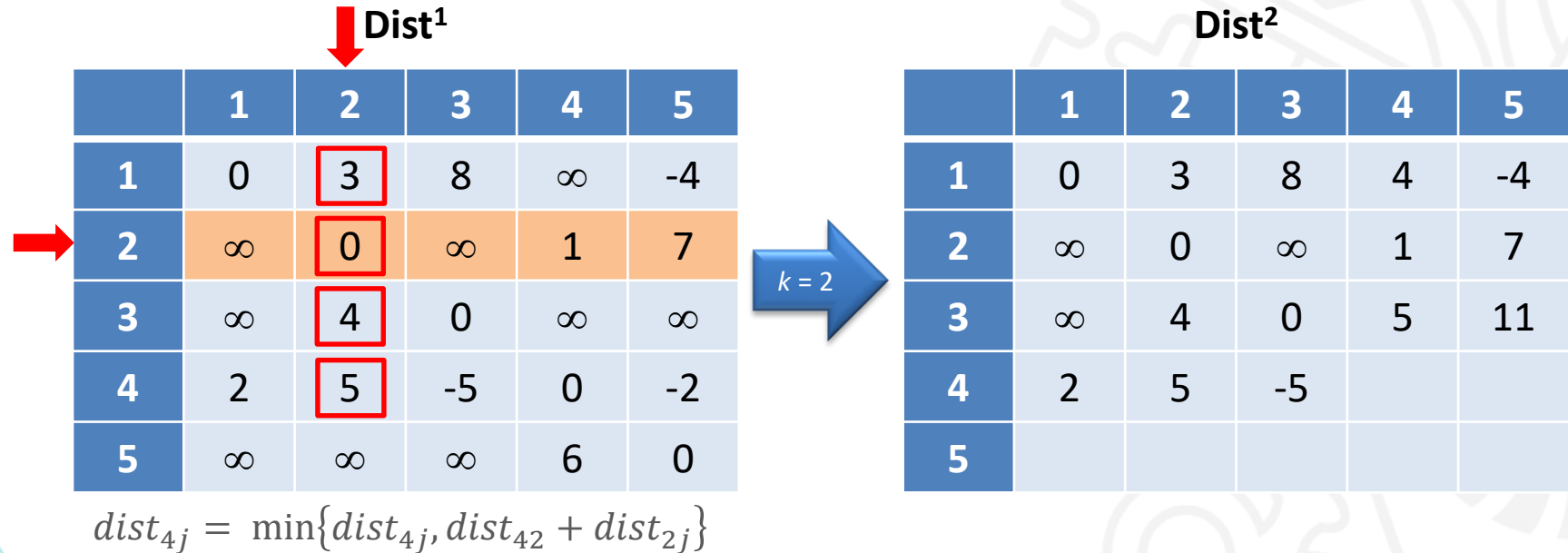
Método de Floyd-Warshall – Exemplo



Método de Floyd-Warshall – Exemplo



Método de Floyd-Warshall – Exemplo



Método de Floyd-Warshall – Exemplo

Dist¹

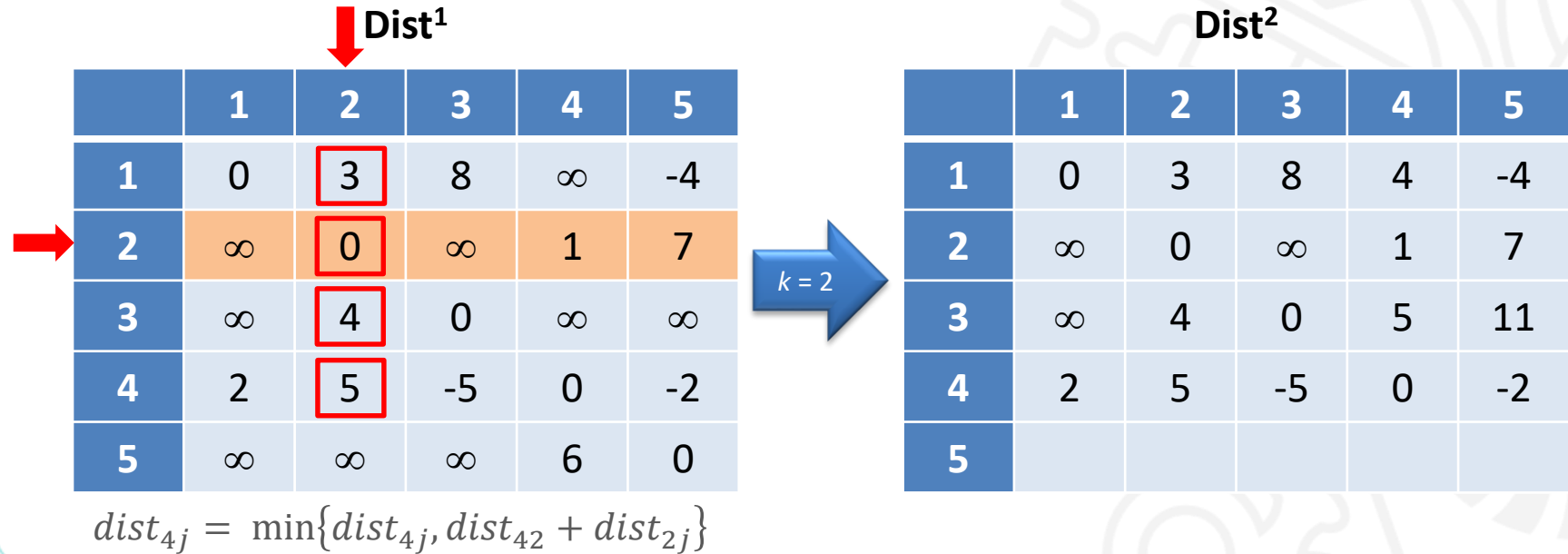
	1	2	3	4	5
1	0	3	8	∞	-4
2	∞	0	∞	1	7
3	∞	4	0	∞	∞
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{4j} = \min\{dist_{4j}, dist_{42} + dist_{2j}\}$$

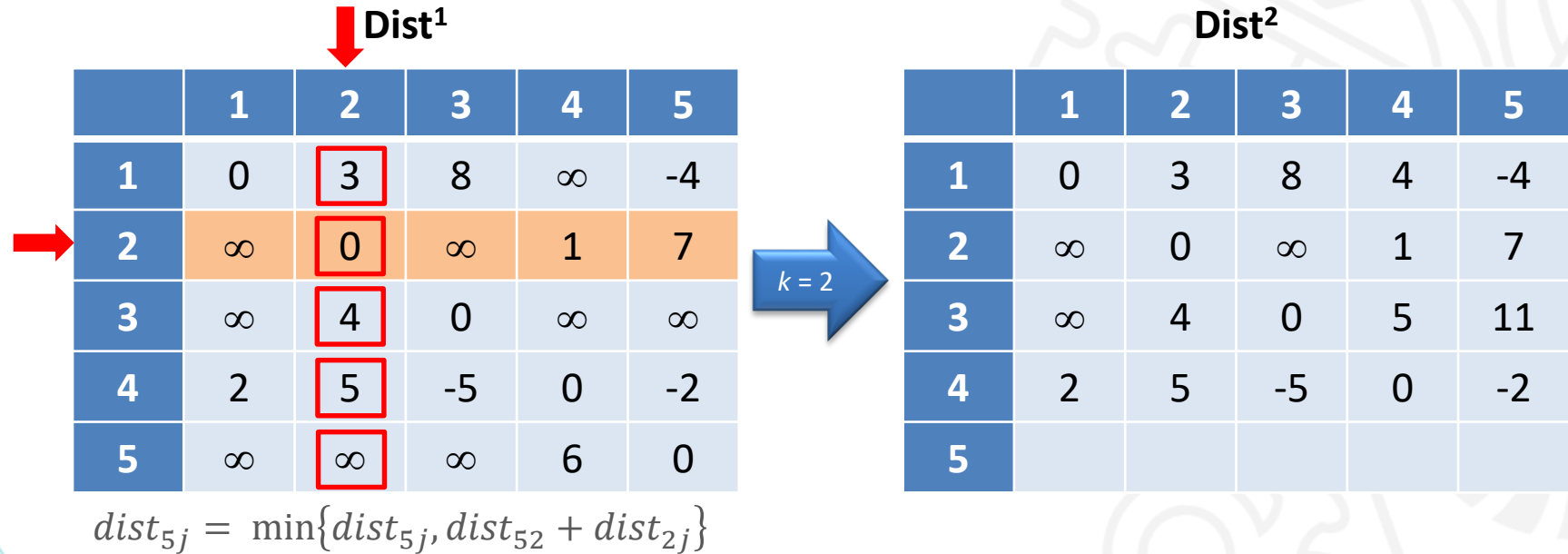
 $k = 2$

	Dist ²				
	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	
5					

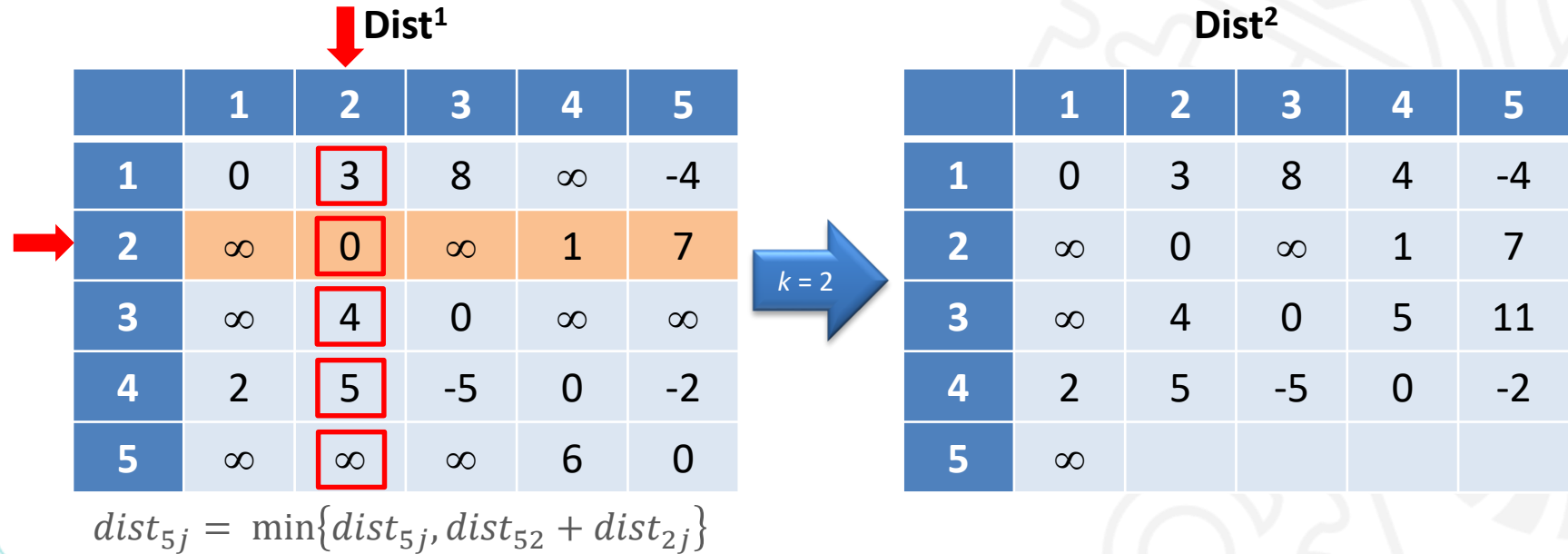
Método de Floyd-Warshall – Exemplo



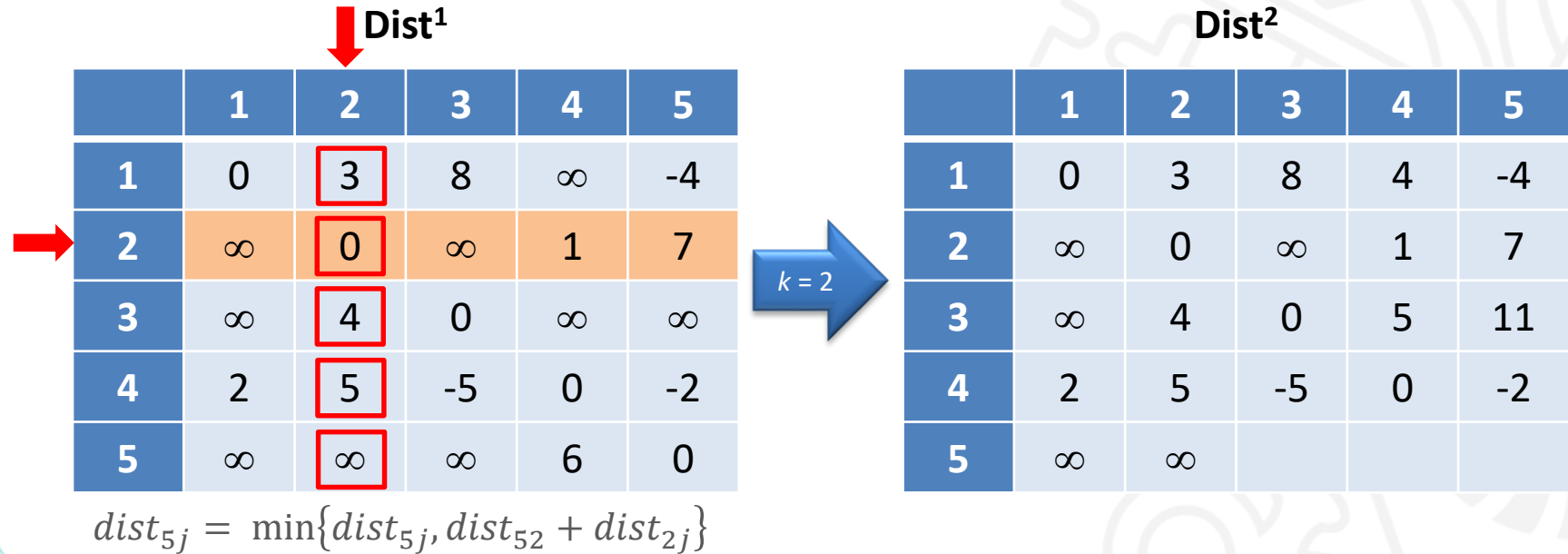
Método de Floyd-Warshall – Exemplo



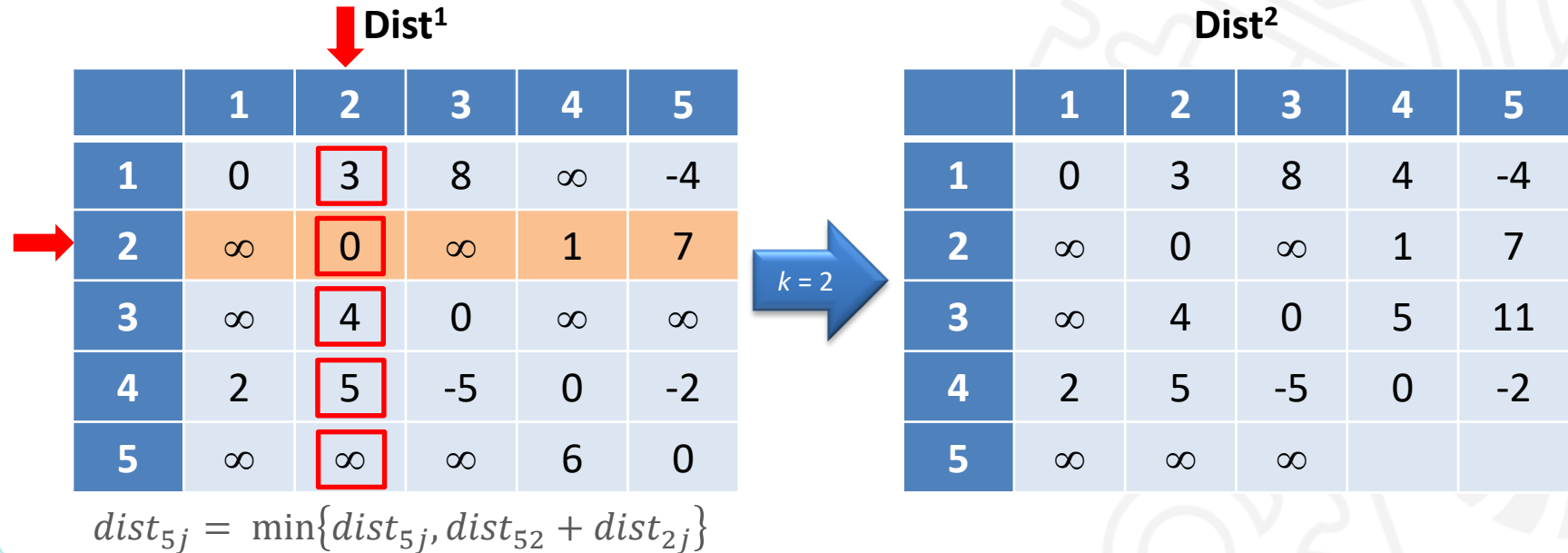
Método de Floyd-Warshall – Exemplo



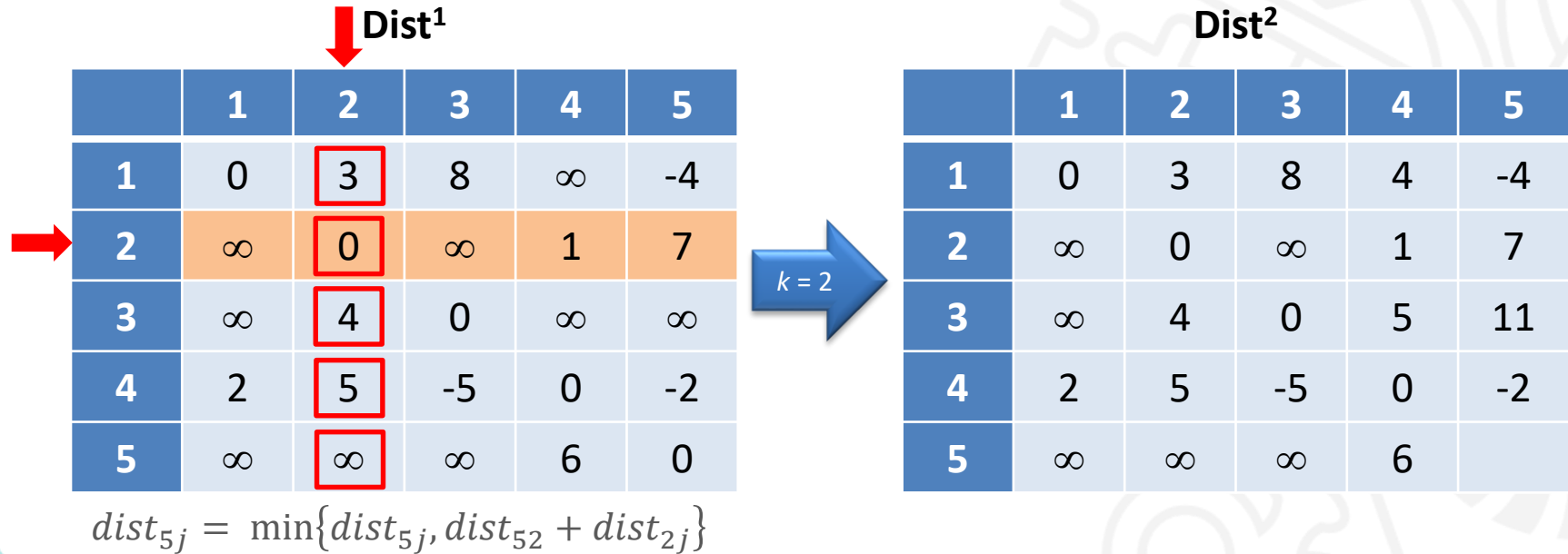
Método de Floyd-Warshall – Exemplo



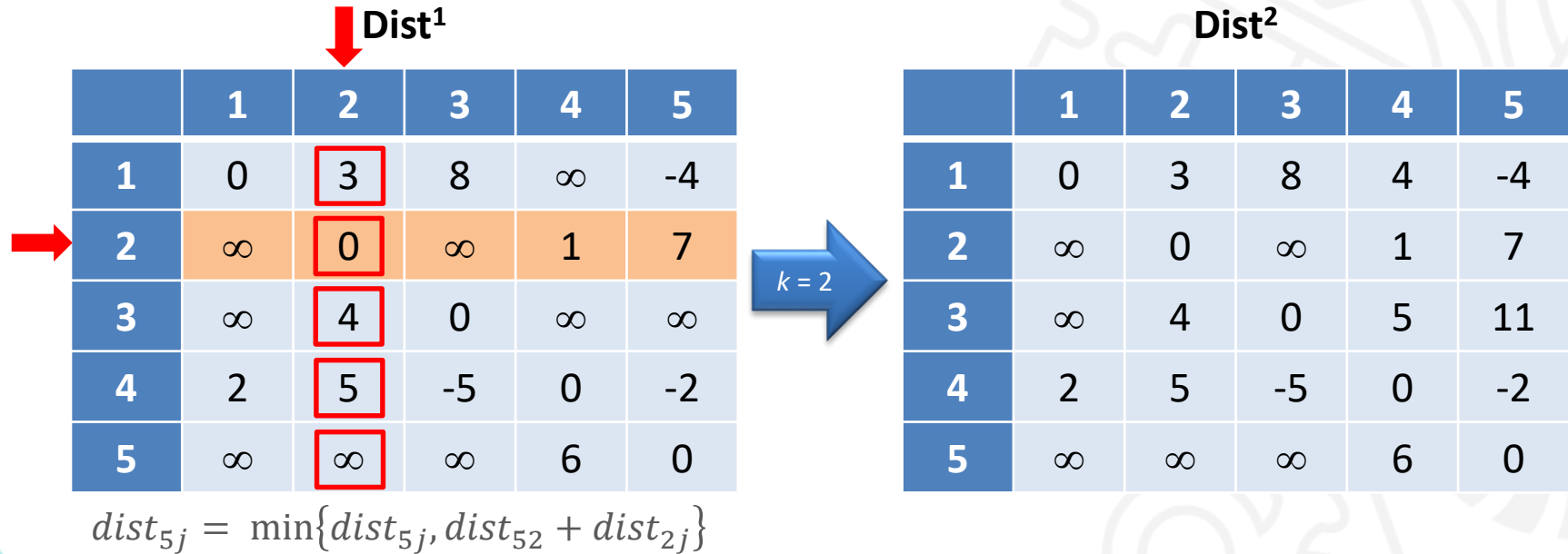
Método de Floyd-Warshall – Exemplo



Método de Floyd-Warshall – Exemplo



Método de Floyd-Warshall – Exemplo



Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

k = 3

Dist³

	1	2	3	4	5
1					
2					
3					
4					
5					

$$dist_{ij} = \min\{dist_{ij}, dist_{i3} + dist_{3j}\}$$

Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{1j} = \min\{dist_{1j}, dist_{13} + dist_{3j}\}$$

$k = 3$

Dist³

	1	2	3	4	5
1					
2					
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{1j} = \min\{dist_{1j}, dist_{13} + dist_{3j}\}$$

$k = 3$

Dist³

	1	2	3	4	5
1	0				
2					
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{1j} = \min\{dist_{1j}, dist_{13} + dist_{3j}\}$$

$k = 3$

Dist³

	1	2	3	4	5
1	0	3			
2					
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{1j} = \min\{dist_{1j}, dist_{13} + dist_{3j}\}$$

$k = 3$

Dist³

	1	2	3	4	5
1	0	3	8		
2					
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{1j} = \min\{dist_{1j}, dist_{13} + dist_{3j}\}$$

k = 3

Dist³

	1	2	3	4	5
1	0	3	8	4	
2					
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{1j} = \min\{dist_{1j}, dist_{13} + dist_{3j}\}$$

k = 3

Dist³

	1	2	3	4	5
1	0	3	8	4	-4
2					
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{2j} = \min\{dist_{2j}, dist_{23} + dist_{3j}\}$$

k = 3

Dist³

	1	2	3	4	5
1	0	3	8	4	-4
2					
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

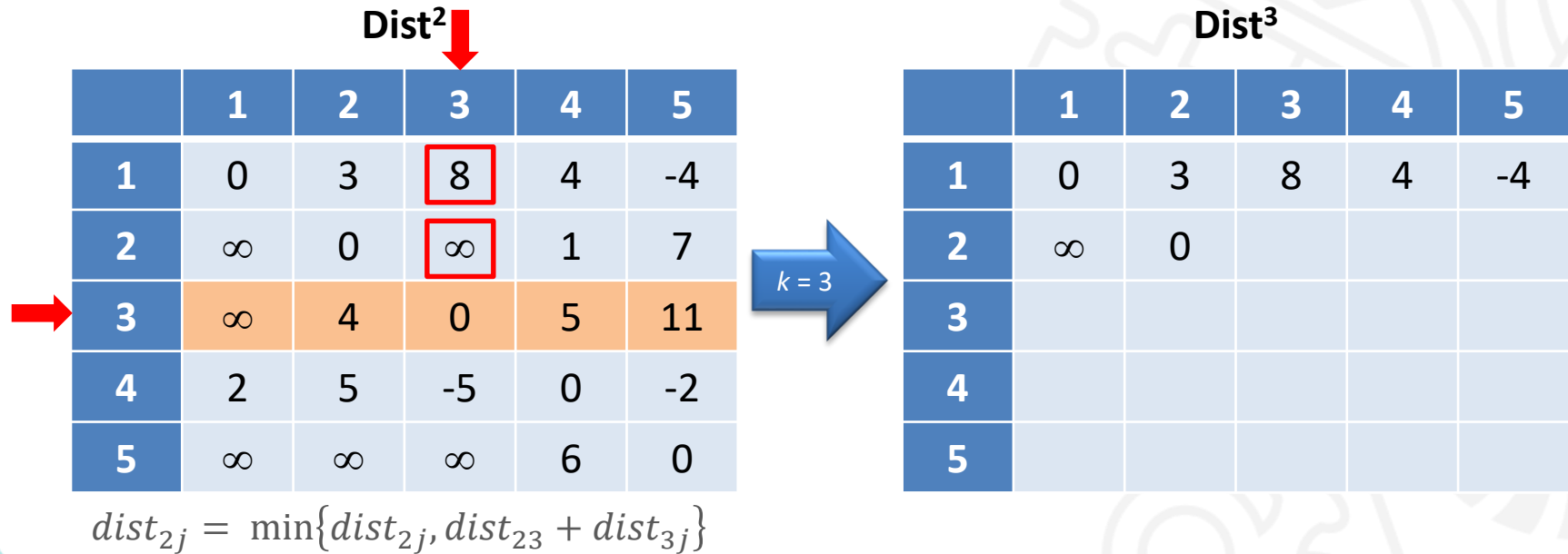
$$dist_{2j} = \min\{dist_{2j}, dist_{23} + dist_{3j}\}$$

k = 3

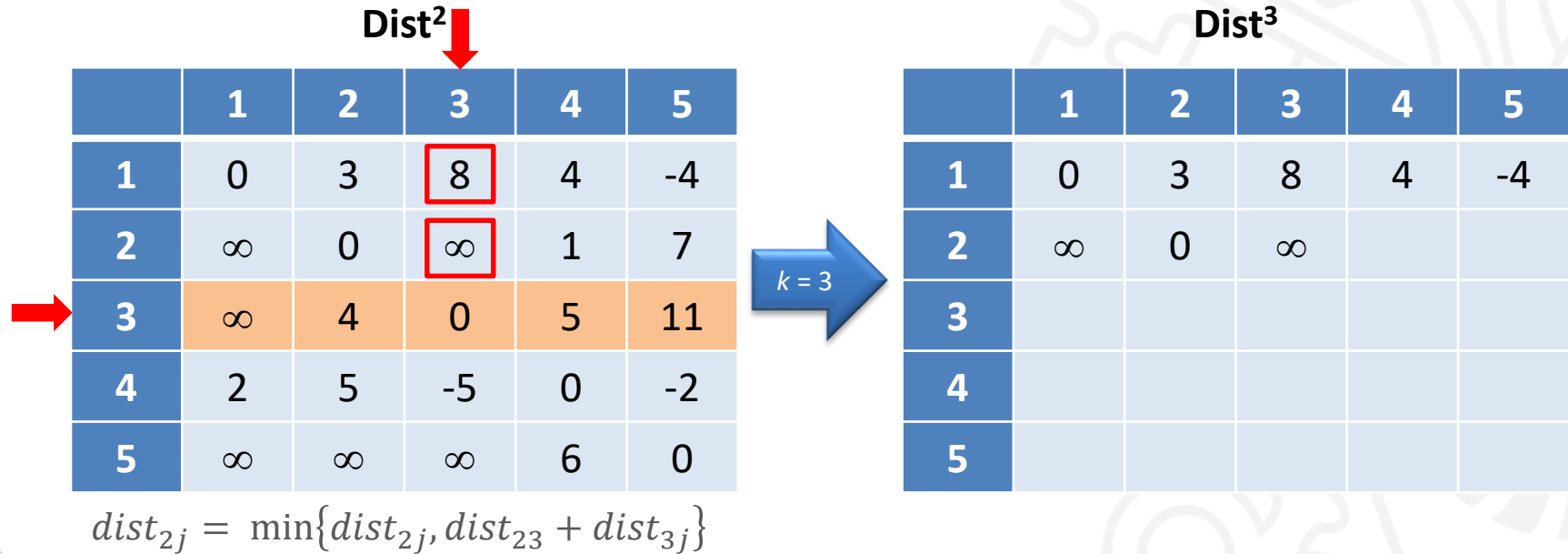
Dist³

	1	2	3	4	5
1	0	3	8	4	-4
2	∞				
3					
4					
5					

Método de Floyd-Warshall – Exemplo



Método de Floyd-Warshall – Exemplo



Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

k = 3

Dist³

	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	
3					
4					
5					

$$dist_{2j} = \min\{dist_{2j}, dist_{23} + dist_{3j}\}$$

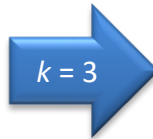
Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{2j} = \min\{dist_{2j}, dist_{23} + dist_{3j}\}$$



Dist³

	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

k = 3

Dist³

	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3					
4					
5					

$$dist_{3j} = \min\{dist_{3j}, dist_{33} + dist_{3j}\}$$

Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

k = 3

Dist³

	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞				
4					
5					

$$dist_{3j} = \min\{dist_{3j}, dist_{33} + dist_{3j}\}$$

Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

$k = 3$

Dist³

	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4			
4					
5					

$$dist_{3j} = \min\{dist_{3j}, dist_{33} + dist_{3j}\}$$

Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

k = 3

Dist³

	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0		
4					
5					

$$dist_{3j} = \min\{dist_{3j}, dist_{33} + dist_{3j}\}$$

Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

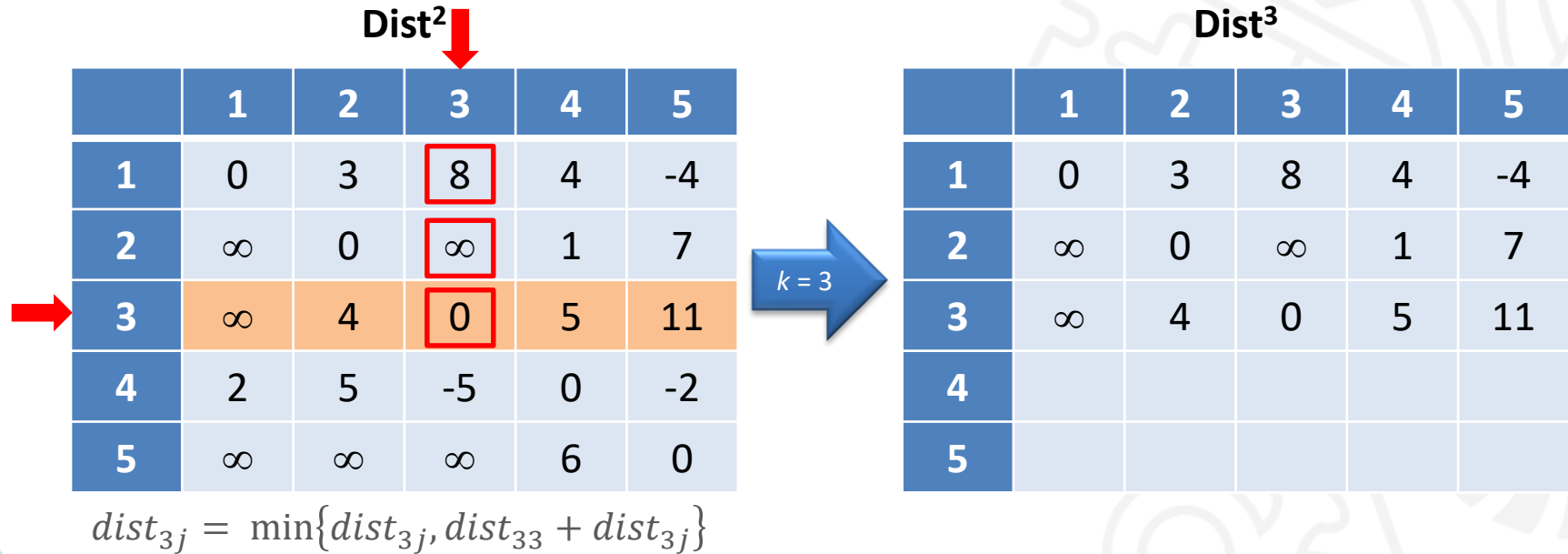
k = 3

Dist³

	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	
4					
5					

$$dist_{3j} = \min\{dist_{3j}, dist_{33} + dist_{3j}\}$$

Método de Floyd-Warshall – Exemplo



Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

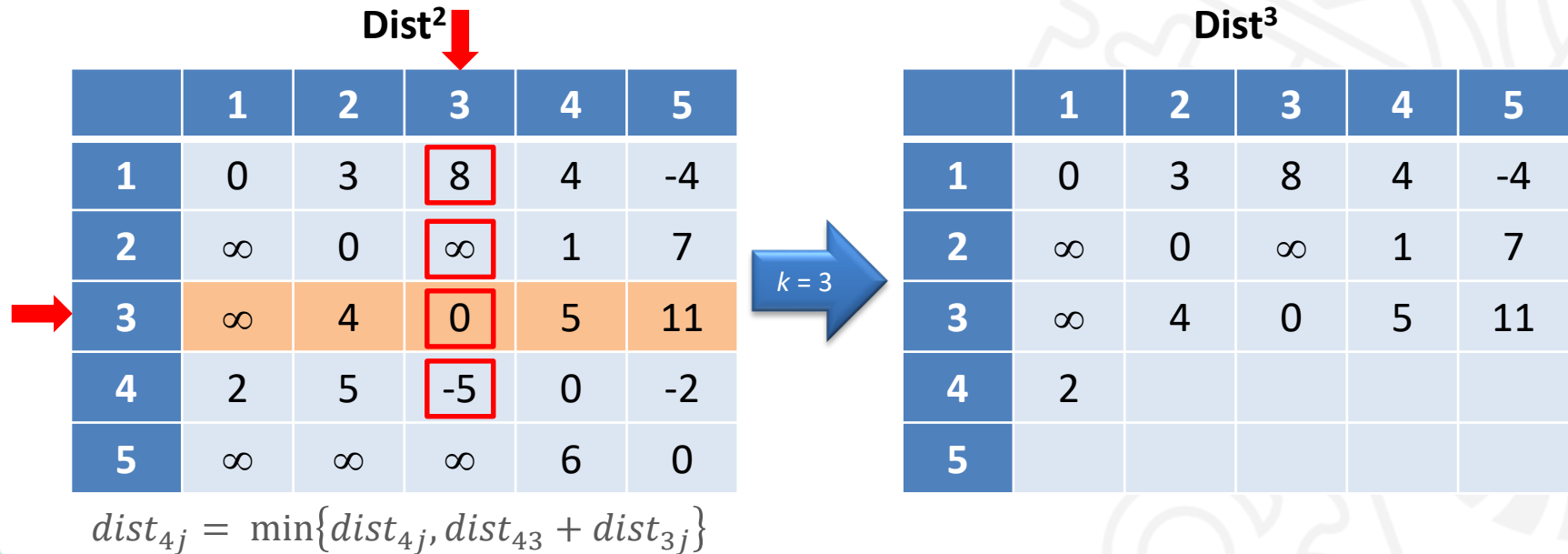
$k = 3$

Dist³

	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4					
5					

$$dist_{4j} = \min\{dist_{4j}, dist_{43} + dist_{3j}\}$$

Método de Floyd-Warshall – Exemplo



Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

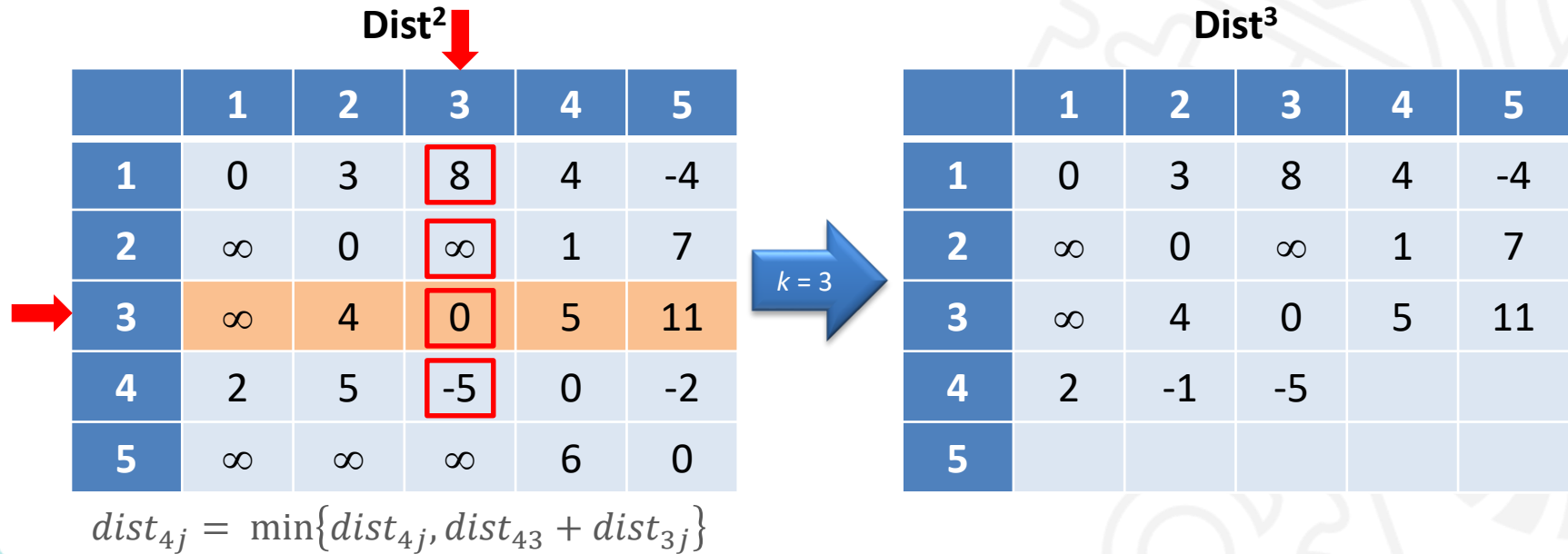
$k = 3$

Dist³

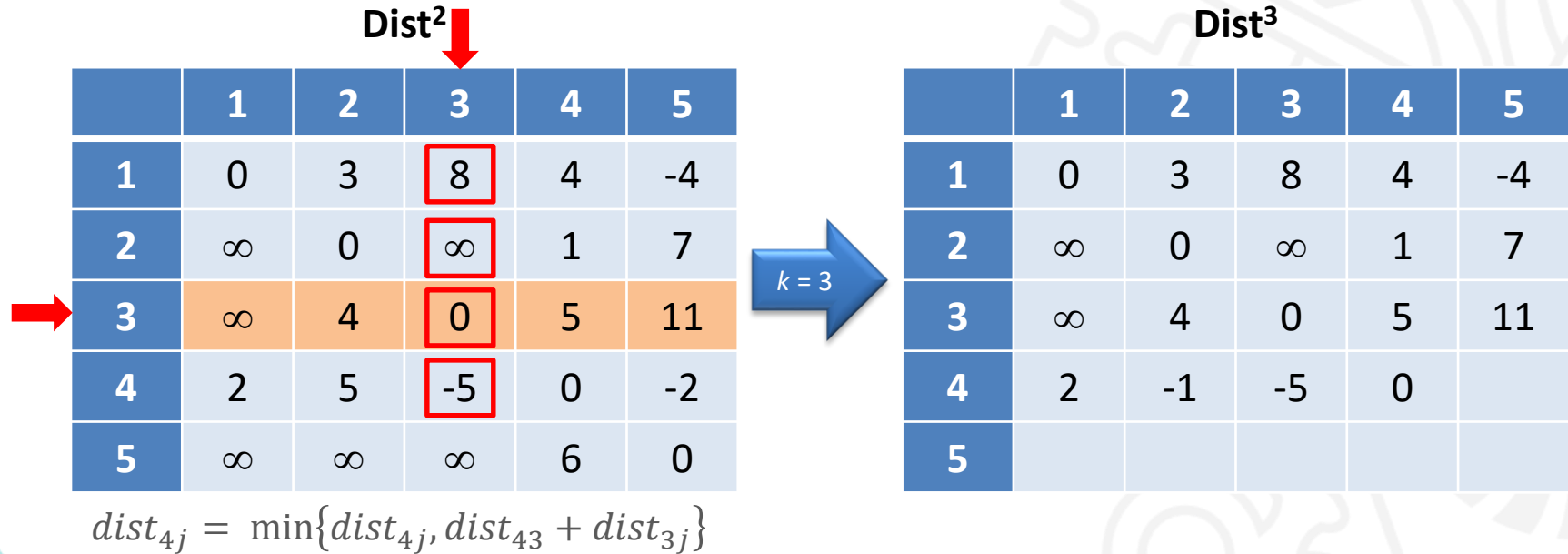
	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1			
5					

$$dist_{4j} = \min\{dist_{4j}, dist_{43} + dist_{3j}\}$$

Método de Floyd-Warshall – Exemplo



Método de Floyd-Warshall – Exemplo



Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

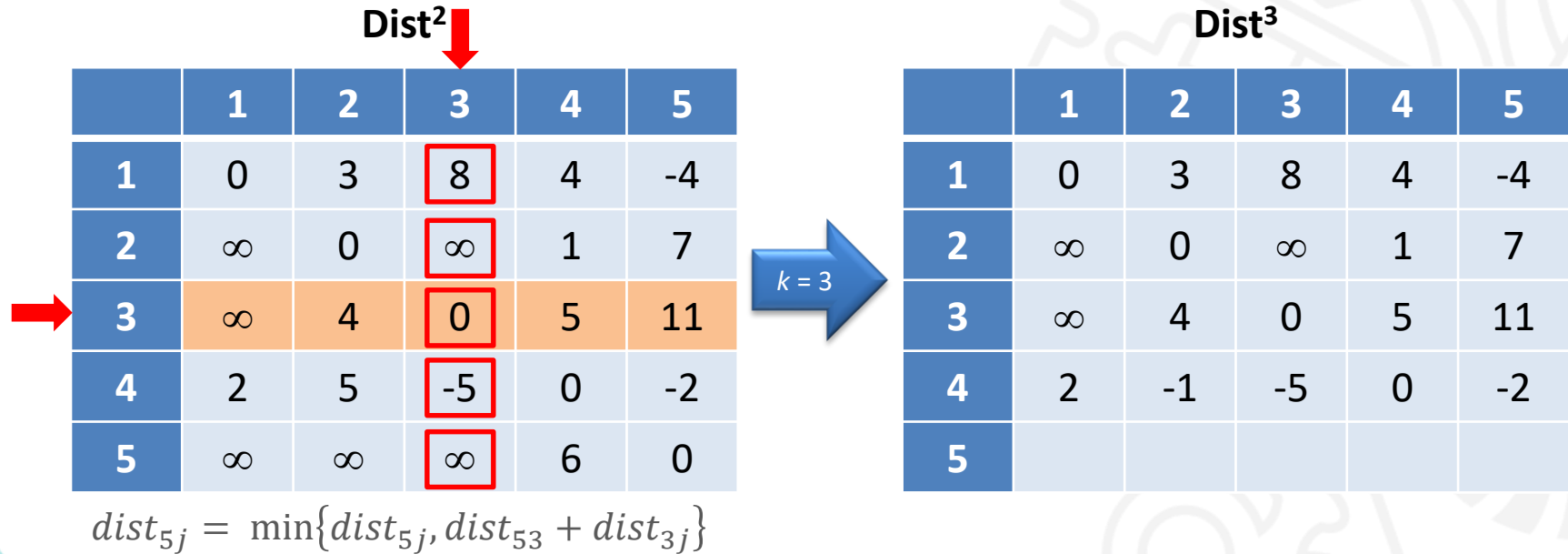
$k = 3$

Dist³

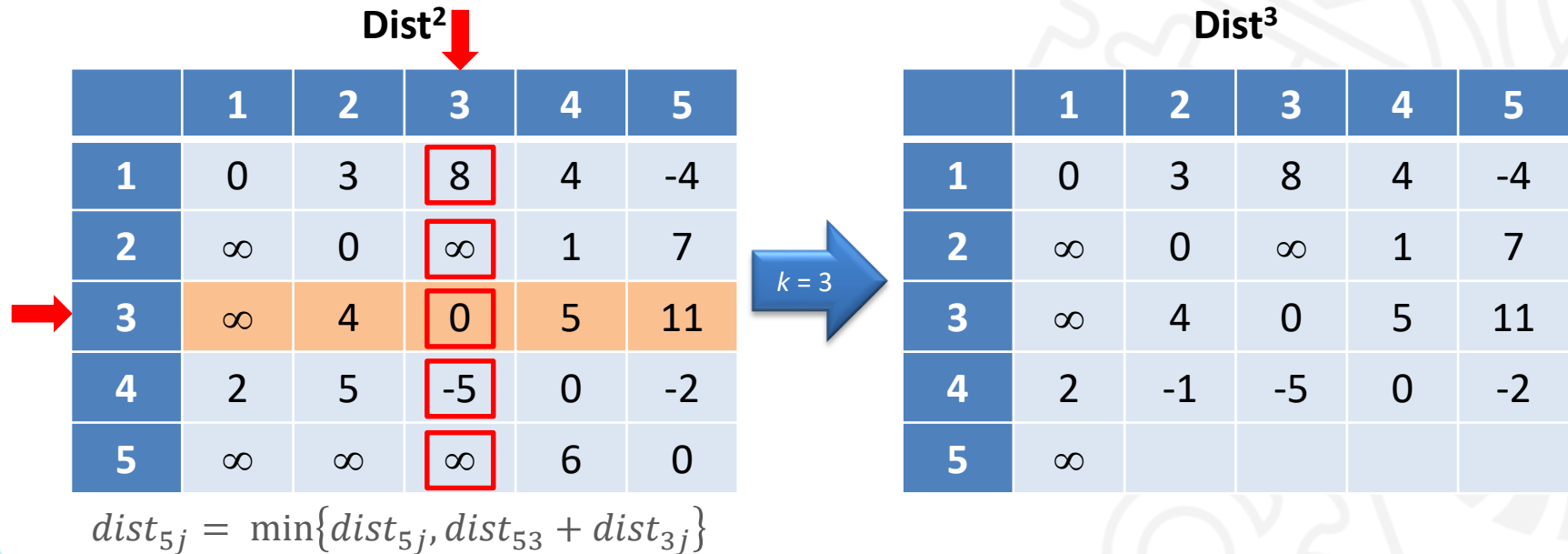
	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5					

$$dist_{4j} = \min\{dist_{4j}, dist_{43} + dist_{3j}\}$$

Método de Floyd-Warshall – Exemplo



Método de Floyd-Warshall – Exemplo



Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

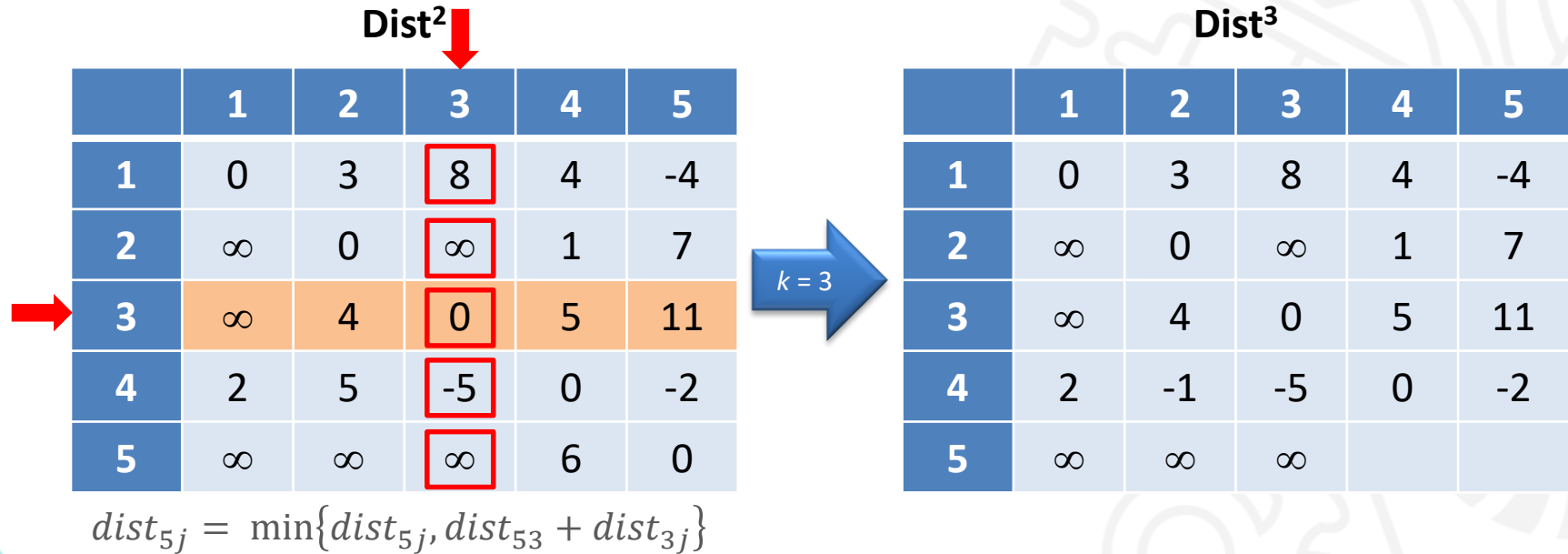
$k = 3$

Dist³

	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞			

$$dist_{5j} = \min\{dist_{5j}, dist_{53} + dist_{3j}\}$$

Método de Floyd-Warshall – Exemplo



Método de Floyd-Warshall – Exemplo

Dist²



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

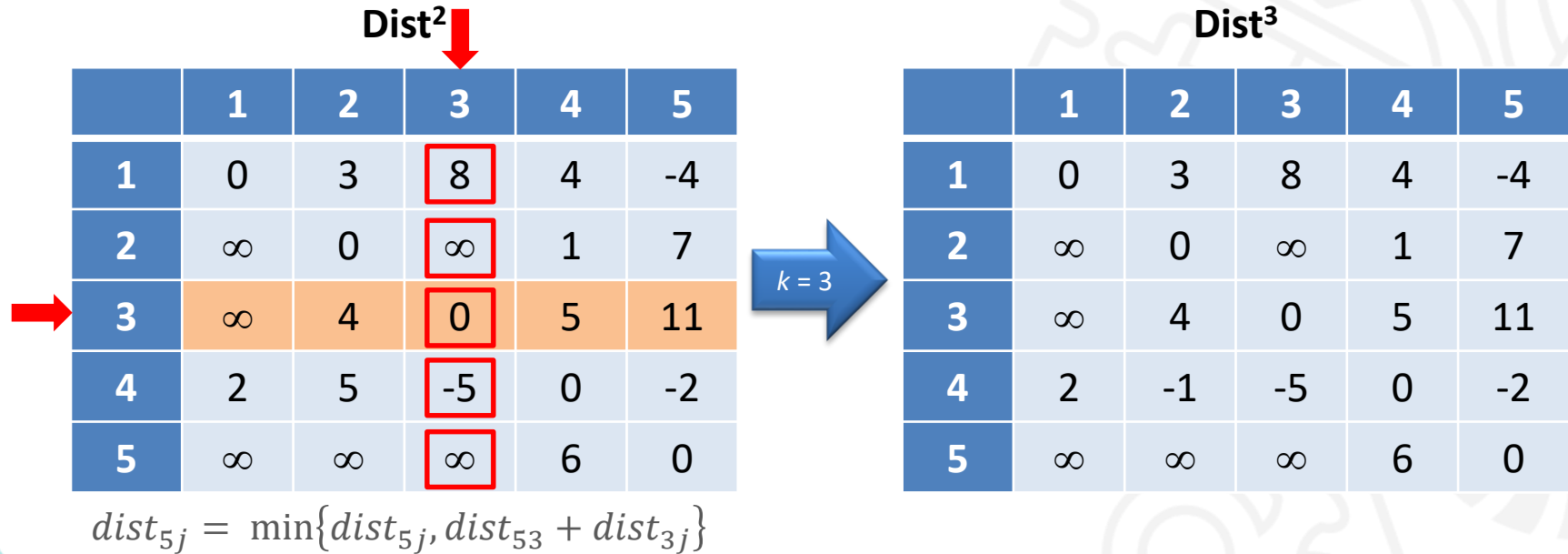
$k = 3$

Dist³

	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	

$$dist_{5j} = \min\{dist_{5j}, dist_{53} + dist_{3j}\}$$

Método de Floyd-Warshall – Exemplo

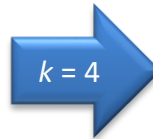


Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0



Dist⁴

	1	2	3	4	5
1					
2					
3					
4					
5					

$$dist_{ij} = \min\{dist_{ij}, dist_{i4} + dist_{4j}\}$$

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0



$$dist_{1j} = \min\{dist_{1j}, dist_{14} + dist_{4j}\}$$

$k = 4$

Dist⁴

	1	2	3	4	5
1					
2					
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0



$$dist_{1j} = \min\{dist_{1j}, dist_{14} + dist_{4j}\}$$

$k = 4$

Dist⁴

	1	2	3	4	5
1	0				
2					
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0



$$dist_{1j} = \min\{dist_{1j}, dist_{14} + dist_{4j}\}$$

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3			
2					
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0



$$dist_{1j} = \min\{dist_{1j}, dist_{14} + dist_{4j}\}$$

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1		
2					
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0



$$dist_{1j} = \min\{dist_{1j}, dist_{14} + dist_{4j}\}$$

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	
2					
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0



$$dist_{1j} = \min\{dist_{1j}, dist_{14} + dist_{4j}\}$$

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2					
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{2j} = \min\{dist_{2j}, dist_{24} + dist_{4j}\}$$

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2					
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{2j} = \min\{dist_{2j}, dist_{24} + dist_{4j}\}$$

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3				
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0



$$dist_{2j} = \min\{dist_{2j}, dist_{24} + dist_{4j}\}$$

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0			
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{2j} = \min\{dist_{2j}, dist_{24} + dist_{4j}\}$$



Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4		
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{2j} = \min\{dist_{2j}, dist_{24} + dist_{4j}\}$$

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0



$$dist_{2j} = \min\{dist_{2j}, dist_{24} + dist_{4j}\}$$

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{3j} = \min\{dist_{3j}, dist_{34} + dist_{4j}\}$$

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7				
4					
5					

$$dist_{3j} = \min\{dist_{3j}, dist_{34} + dist_{4j}\}$$

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{3j} = \min\{dist_{3j}, dist_{34} + dist_{4j}\}$$

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4			
4					
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{3j} = \min\{dist_{3j}, dist_{34} + dist_{4j}\}$$

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0		
4					
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$$dist_{3j} = \min\{dist_{3j}, dist_{34} + dist_{4j}\}$$

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	
4					
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4					
5					

$$dist_{3j} = \min\{dist_{3j}, dist_{34} + dist_{4j}\}$$

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4					
5					

$$dist_{4j} = \min\{dist_{4j}, dist_{44} + dist_{4j}\}$$

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2				
5					

$$dist_{4j} = \min\{dist_{4j}, dist_{44} + dist_{4j}\}$$

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1			
5					

$$dist_{4j} = \min\{dist_{4j}, dist_{44} + dist_{4j}\}$$

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0



$$dist_{4j} = \min\{dist_{4j}, dist_{44} + dist_{4j}\}$$

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5		
5					

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	
5					

$$dist_{4j} = \min\{dist_{4j}, dist_{44} + dist_{4j}\}$$

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5					

$$dist_{4j} = \min\{dist_{4j}, dist_{44} + dist_{4j}\}$$

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5					

$$dist_{5j} = \min\{dist_{5j}, dist_{54} + dist_{4j}\}$$

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8				

$$dist_{5j} = \min\{dist_{5j}, dist_{54} + dist_{4j}\}$$

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0

$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5			

$$dist_{5j} = \min\{dist_{5j}, dist_{54} + dist_{4j}\}$$

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0



$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1		

$$dist_{5j} = \min\{dist_{5j}, dist_{54} + dist_{4j}\}$$

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0



$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	

$$dist_{5j} = \min\{dist_{5j}, dist_{54} + dist_{4j}\}$$

Método de Floyd-Warshall – Exemplo

Dist³



	1	2	3	4	5
1	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	-1	-5	0	-2
5	∞	∞	∞	6	0



$k = 4$

Dist⁴

	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$$dist_{5j} = \min\{dist_{5j}, dist_{54} + dist_{4j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$k = 5$

Dist⁵

	1	2	3	4	5
1					
2					
3					
4					
5					

$$dist_{ij} = \min\{dist_{ij}, dist_{i5} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$$dist_{1j} = \min\{dist_{1j}, dist_{15} + dist_{5j}\}$$



Dist⁵

	1	2	3	4	5
1					
2					
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$k = 5$

Dist⁵

	1	2	3	4	5
1	0				
2					
3					
4					
5					

$$dist_{1j} = \min\{dist_{1j}, dist_{15} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0



$$dist_{1j} = \min\{dist_{1j}, dist_{15} + dist_{5j}\}$$



Dist⁵

	1	2	3	4	5
1	0	1			
2					
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3		
2					
3					
4					
5					

$$dist_{1j} = \min\{dist_{1j}, dist_{15} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	
2					
3					
4					
5					

$$dist_{1j} = \min\{dist_{1j}, dist_{15} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2					
3					
4					
5					

$$dist_{1j} = \min\{dist_{1j}, dist_{15} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2					
3					
4					
5					

$$dist_{2j} = \min\{dist_{2j}, dist_{25} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3				
3					
4					
5					

$$dist_{2j} = \min\{dist_{2j}, dist_{25} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0



$$dist_{2j} = \min\{dist_{2j}, dist_{25} + dist_{5j}\}$$

$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0			
3					
4					
5					

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0



$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4		
3					
4					
5					

$$dist_{2j} = \min\{dist_{2j}, dist_{25} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	
3					
4					
5					

$$dist_{2j} = \min\{dist_{2j}, dist_{25} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0



$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3					
4					
5					

$$dist_{2j} = \min\{dist_{2j}, dist_{25} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0



$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3					
4					
5					

$$dist_{3j} = \min\{dist_{3j}, dist_{35} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0



$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7				
4					
5					

$$dist_{3j} = \min\{dist_{3j}, dist_{35} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0



$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4			
4					
5					

$$dist_{3j} = \min\{dist_{3j}, dist_{35} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0



$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0		
4					
5					

$$dist_{3j} = \min\{dist_{3j}, dist_{35} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0



$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	
4					
5					

$$dist_{3j} = \min\{dist_{3j}, dist_{35} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0



$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4					
5					

$$dist_{3j} = \min\{dist_{3j}, dist_{35} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4					
5					

$$dist_{4j} = \min\{dist_{4j}, dist_{45} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2				
5					

$$dist_{4j} = \min\{dist_{4j}, dist_{45} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0



$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1			
5					

$$dist_{4j} = \min\{dist_{4j}, dist_{45} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5		
5					

$$dist_{4j} = \min\{dist_{4j}, dist_{45} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	
5					

$$dist_{4j} = \min\{dist_{4j}, dist_{45} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5					

$$dist_{4j} = \min\{dist_{4j}, dist_{45} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5					

$$dist_{5j} = \min\{dist_{5j}, dist_{55} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8				

$$dist_{5j} = \min\{dist_{5j}, dist_{55} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5			

$$dist_{5j} = \min\{dist_{5j}, dist_{55} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0



$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1		

$$dist_{5j} = \min\{dist_{5j}, dist_{55} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0



$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	

$$dist_{5j} = \min\{dist_{5j}, dist_{55} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo

Dist⁴



	1	2	3	4	5
1	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

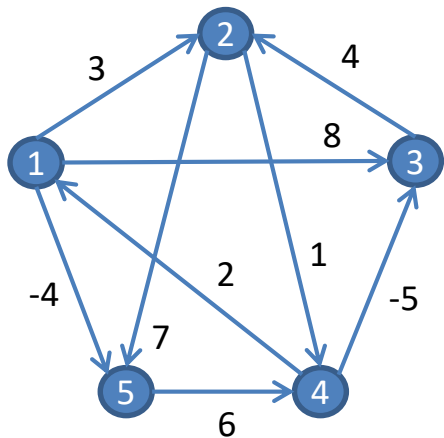
$k = 5$

Dist⁵

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

$$dist_{5j} = \min\{dist_{5j}, dist_{55} + dist_{5j}\}$$

Método de Floyd-Warshall – Exemplo



Distâncias Mínimas

	1	2	3	4	5
1	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

